

Transformer是样本高效的世界模型

Vincent Micheli*日内瓦大学

Eloi Alonso*日内瓦大学

François Fleuret日内瓦大学

摘要

深度强化学习智能体以样本效率低下而闻名，这极大地限制了其在现实问题中的应用。近来，许多基于模型的方法被设计出来以解决这一问题，其中在由世界模型模拟的想象中学习是最为突出的方法之一。然而，尽管与模拟环境进行几乎无限的交互听起来很诱人，但世界模型必须在长时间跨度内保持准确性。受Transformer在序列建模任务中取得成功的启发，我们提出了IRIS，这是一种在由离散自编码器和自回归Transformer组成的世界模型中学习的数据高效智能体。在Atari 100k基准中， 仅需相当于两小时的游戏时间，IRIS就达到了1.046的平均人类归一化得分，在26款游戏中于10款上超越了人类表现，为无需前瞻搜索的方法树立了新的最先进水平。为了促进未来关于Transformer和世界模型在样本高效强化学习中的研究，我们在<https://github.com/eloialonso/iris>上发布了我们的代码和模型。

1 引言

深度强化学习已成为在具有挑战性的环境中培养胜任智能体的主导范式。最值得注意的是，深度强化学习算法在大量街机游戏（Mnih 等人，2015；Schrittwieser等人，2020；Hafner 等人，2021）、即时战略游戏（Vinyals 等人，2019；Berner 等人，2019）、棋盘游戏（Silver 等人，2016；2018；Schrittwieser等人，2020）以及不完全信息游戏（Schmid 等人，2021；Brown等，2020a）中取得了令人瞩目的性能。然而，这些方法的一个共同缺点是样本效率极低。事实上，经验需求范围从DreamerV2（Hafner 等人，2021）在Atari 2600游戏（Bellemare 等人，2013b）中所需的数月游戏时间，到OpenAI Five在Dota2（Berner 等人，2019）中所需的数千年游戏时间不等。虽然某些环境可以加速训练智能体，但现实世界的应用通常无法做到。此外，与环境交互次数相关的额外成本或安全考虑也可能出现（Yampolskiy，2018）。因此，样本效率是弥合研究差距与部署深度强化学习智能体于现实环境之间的必要条件。

基于模型的方法（Sutton & Barto, 2018）是实现数据高效性的重要方向。近期，世界模型通过多种方式得到利用：纯表示学习（Schwarzer等人，2021）、前瞻搜索（Schrittwieser等人，2020；Ye 等人，2021）以及在想象中学习（Ha & Schmidhuber，2018；Kaiser等人，2020；Hafner 等人，2020；2021）。后一种方法尤为吸引人，因为在世界模型内部训练智能体可使其摆脱样本效率的限制。然而，该框架高度依赖准确的世界模型，因为策略纯粹在想象中训练。在早期开创性工作中，Ha & Schmidhuber（2018）成功地在玩具环境中构建了基于想象的智能体。SimPLe最近在更具挑战性的Atari 100k基准（Kaiser等人，2020）中展现出潜力。目前，在想象中学习的最佳Atari智能体是DreamerV2（Hafner 等人，2021），尽管其开发和评估是在在两亿帧数据的支持下完成的，远未达到样本高效的区域。因此，设计能够以少量样本处理视觉复杂且部分可观测环境的新型世界模型架构，是发挥其作为替代训练场所潜力的关键。

Transformer架构（Vaswani等，2017）如今在自然语言处理领域（Devlin等，2019；Radford等，2019；Brown等，2020b；Raffel等，2020）无处不在，同时也在计算机视觉领域（Dosovitskiy等，2021；He等，2022）以及离线

*同等贡献，排名由抛硬币决定。通讯作者：{first.last}@unige.ch

TRANSFORMERS ARE SAMPLE-EFFICIENT WORLD MODELS

Vincent Micheli*University of Geneva

Eloi Alonso*University of Geneva

François FleuretUniversity of Geneva

ABSTRACT

Deep reinforcement learning agents are notoriously sample inefficient, which considerably limits their application to real-world problems. Recently, many model-based methods have been designed to address this issue, with learning in the imagination of a world model being one of the most prominent approaches. However, while virtually unlimited interaction with a simulated environment sounds appealing, the world model has to be accurate over extended periods of time. Motivated by the success of Transformers in sequence modeling tasks, we introduce IRIS, a data-efficient agent that learns in a world model composed of a discrete autoencoder and an autoregressive Transformer. With the equivalent of only two hours of gameplay in the Atari 100k benchmark, IRIS achieves a mean human normalized score of 1.046, and outperforms humans on 10 out of 26 games, setting a new state of the art for methods without lookahead search. To foster future research on Transformers and world models for sample-efficient reinforcement learning, we release our code and models at <https://github.com/eloialonso/iris>.

1 INTRODUCTION

Deep Reinforcement Learning (RL) has become the dominant paradigm for developing competent agents in challenging environments. Most notably, deep RL algorithms have achieved impressive performance in a multitude of arcade (Mnih et al., 2015; Schrittwieser et al., 2020; Hafner et al., 2021), real-time strategy (Vinyals et al., 2019; Berner et al., 2019), board (Silver et al., 2016; 2018; Schrittwieser et al., 2020) and imperfect information (Schmid et al., 2021; Brown et al., 2020a) games. However, a common drawback of these methods is their extremely low sample efficiency. Indeed, experience requirements range from months of gameplay for DreamerV2 (Hafner et al., 2021) in Atari 2600 games (Bellemare et al., 2013b) to thousands of years for OpenAI Five in Dota2 (Berner et al., 2019). While some environments can be sped up for training agents, real-world applications often cannot. Besides, additional cost or safety considerations related to the number of environmental interactions may arise (Yampolskiy, 2018). Hence, sample efficiency is a necessary condition to bridge the gap between research and the deployment of deep RL agents in the wild.

Model-based methods (Sutton & Barto, 2018) constitute a promising direction towards data efficiency. Recently, world models were leveraged in several ways: pure representation learning (Schwarzer et al., 2021), lookahead search (Schrittwieser et al., 2020; Ye et al., 2021), and learning in imagination (Ha & Schmidhuber, 2018; Kaiser et al., 2020; Hafner et al., 2020; 2021). The latter approach is particularly appealing because training an agent inside a world model frees it from sample efficiency constraints. Nevertheless, this framework relies heavily on accurate world models since the policy is purely trained in imagination. In a pioneering work, Ha & Schmidhuber (2018) successfully built imagination-based agents in toy environments. SimPLe recently showed promise in the more challenging Atari 100k benchmark (Kaiser et al., 2020). Currently, the best Atari agent learning in imagination is DreamerV2 (Hafner et al., 2021), although it was developed and evaluated with two hundred million frames available, far from the sample-efficient regime. Therefore, designing new world model architectures, capable of handling visually complex and partially observable environments with few samples, is key to realize their potential as surrogate training grounds.

The Transformer architecture (Vaswani et al., 2017) is now ubiquitous in Natural Language Processing (Devlin et al., 2019; Radford et al., 2019; Brown et al., 2020b; Raffel et al., 2020), and is also gaining traction in Computer Vision (Dosovitskiy et al., 2021; He et al., 2022), as well as in Offline

*Equal contributions, order determined by a coin flip. Correspondence: {first.last}@unige.ch

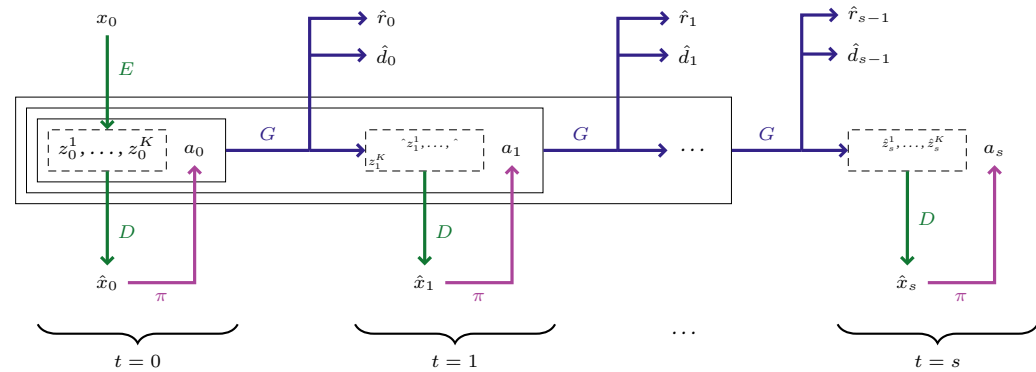


图1: 随时间展开的想象过程。此图展示了策略 π (以紫色箭头表示) 在想象空间中执行一系列动作。绿色箭头对应离散自编码器的编码器 E 和解码器 D ，其任务是用习得的符号语言表示帧。世界模型的主干 G 是一个类GPT Transformer，以蓝色箭头表示。对于策略 π 采取的动作， G 通过自回归方式展开新的帧标记（ D 可解码这些标记）来模拟环境动力学。 G 还预测奖励和潜在的情节终止。更具体地说，初始帧 x_0 由 E 编码为令牌 $z_0 = (z_0^1, \dots, z_0^K) = E(x_0)$ 。解码器 D 重建图像 $\hat{x}_0 = D(z_0)$ ，策略 π 据此预测动作 a_0 。基于 z_0 和 a_0 ， G 预测奖励 $\hat{r}_0$ 、情节终止 $\hat{d}_0 \in \{0, 1\}$ ，并以自回归方式预测下一帧的令牌 $\hat{z}_1 = (\hat{z}_1^1, \dots, \hat{z}_1^K)$ 。虚线框表示给定时间步的图像词元，实线框表示 G 的输入序列，即 z_0 时刻的 $(a_0, t=0)$ 、 z_0 时刻的 $(a_0, z_1, \hat{a}_1, t=1)$ 等。策略 π 完全通过想象轨迹进行训练，仅在实际环境中部署以改进世界模型 (E, D, G) 。

强化学习领域 (Janner 等人, 2021; Chen 等人, 2021) 日益受到重视。特别是GPT系列模型 (Radford 等, 2018; 2019; Brown 等, 2020b) 在语言理解任务中取得了令人瞩目的成果。与世界模型类似，这些基于注意力的模型通过高维信号和自监督学习目标进行训练，因此成为模拟环境的理想候选方案。

Transformer在操作离散标记序列时尤其出色 (Devlin 等, 2019; Brown 等, 2020b)。对于文本数据，有简单的方法 (Schuster & Nakajima, 2012; Kudo & Richardson, 2018) 来构建词汇表，但这种转换对图像而言并不直接。一种朴素的方法是将像素视为图像词元，但标准Transformer架构的计算复杂度与序列长度呈二次方关系，使得这一思路在计算上难以实现。为解决此问题，VQGAN (Esser 等人, 2021) 和DALL-E (Ramesh 等人, 2021) 采用离散自编码器 (Van Den Oord 等人, 2017) 作为从原始像素到数量更少的图像词元的映射。结合自回归Transformer，这些方法展现出强大的无条件与条件图像生成能力。此类结果表明了一种设计世界模型的新途径。

在本文中，我们介绍了 IRIS (基于内部言语的自回归想象)，这是一个在由离散自编码器和自回归Transformer组成的世界模型的想象中训练的智能体。IRIS 通过精确模拟数百万条轨迹来学习行为。我们的方法将动力学学习视为一个序列建模问题，其中自编码器构建图像词元的语言，而Transformer则随时间组合该语言。只需极少的调优，IRIS 便在 Atari 100k 基准 (Kaiser 等人, 2020) 的样本高效强化学习中超越了近期一系列方法 (Kaiser 等人, 2020; Hessel 等人, 2018; Laskin 等人, 2020; Yarats 等人, 2021; Schwarzer 等人, 2021)。仅经过两小时的实时体验，它便达到了 1.046 的平均人类归一化得分，并在 26 款游戏中的 10 款上实现了超人性能。我们在第 2 节中描述 IRIS，并在第 3 节中展示我们的结果。

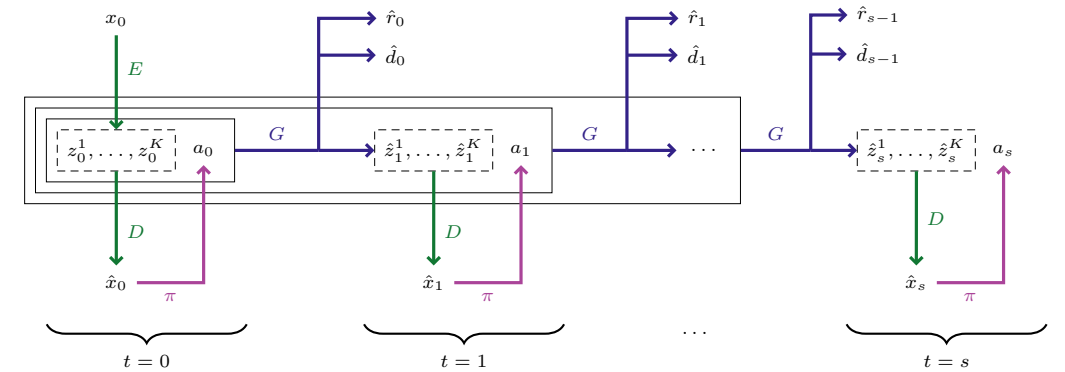


Figure 1: Unrolling imagination over time. This figure shows the policy π , depicted with purple arrows, taking a sequence of actions in imagination. The green arrows correspond to the encoder E and the decoder D of a discrete autoencoder, whose task is to represent frames in its learnt symbolic language. The backbone G of the world model is a GPT-like Transformer, illustrated with blue arrows. For each action that the policy π takes, G simulates the environment dynamics, by autoregressively unfolding new frame tokens that D can decode. G also predicts a reward and a potential episode termination. More specifically, an initial frame x_0 is encoded with E into tokens $z_0 = (z_0^1, \dots, z_0^K) = E(x_0)$. The decoder D reconstructs an image $\hat{x}_0 = D(z_0)$, from which the policy π predicts the action a_0 . From z_0 and a_0 , G predicts the reward $\hat{r}_0$, episode termination $\hat{d}_0 \in \{0, 1\}$, and in an autoregressive manner $\hat{z}_1 = (\hat{z}_1^1, \dots, \hat{z}_1^K)$, the tokens for the next frame. A dashed box indicates image tokens for a given time step, whereas a solid box represents the input sequence of G , i.e. (z_0, a_0) at $t=0$, $(z_0, a_0, \hat{z}_1, a_1)$ at $t=1$, etc. The policy π is purely trained with imagined trajectories, and is only deployed in the real environment to improve the world model (E, D, G) .

Reinforcement Learning (Janner et al., 2021; Chen et al., 2021). In particular, the GPT (Radford et al., 2018; 2019; Brown et al., 2020b) family of models delivered impressive results in language understanding tasks. Similarly to world models, these attention-based models are trained with high-dimensional signals and a self-supervised learning objective, thus constituting ideal candidates to simulate an environment.

Transformers particularly shine when they operate over sequences of discrete tokens (Devlin et al., 2019; Brown et al., 2020b). For textual data, there are simple ways (Schuster & Nakajima, 2012; Kudo & Richardson, 2018) to build a vocabulary, but this conversion is not straightforward with images. A naive approach would consist in treating pixels as image tokens, but standard Transformer architectures scale quadratically with sequence length, making this idea computationally intractable. To address this issue, VQGAN (Esser et al., 2021) and DALL-E (Ramesh et al., 2021) employ a discrete autoencoder (Van Den Oord et al., 2017) as a mapping from raw pixels to a much smaller amount of image tokens. Combined with an autoregressive Transformer, these methods demonstrate strong unconditional and conditional image generation capabilities. Such results suggest a new approach to design world models.

In the present work, we introduce IRIS (Imagination with auto-Regression over an Inner Speech), an agent trained in the imagination of a world model composed of a discrete autoencoder and an autoregressive Transformer. IRIS learns behaviors by accurately simulating millions of trajectories. Our approach casts dynamics learning as a sequence modeling problem, where an autoencoder builds a language of image tokens and a Transformer composes that language over time. With minimal tuning, IRIS outperforms a line of recent methods (Kaiser et al., 2020; Hessel et al., 2018; Laskin et al., 2020; Yarats et al., 2021; Schwarzer et al., 2021) for sample-efficient RL in the Atari 100k benchmark (Kaiser et al., 2020). After only two hours of real-time experience, it achieves a mean human normalized score of 1.046, and reaches superhuman performance on 10 out of 26 games. We describe IRIS in Section 2 and present our results in Section 3.

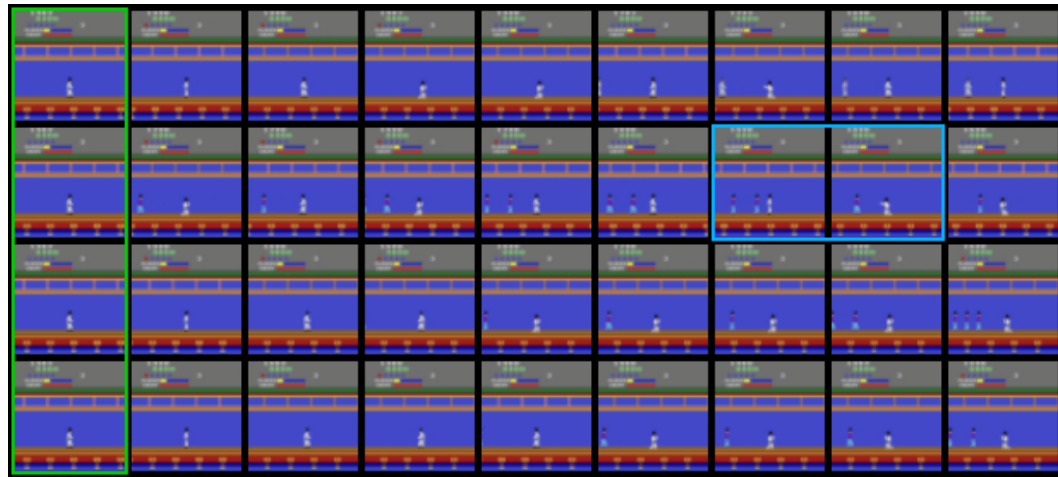


图2: *KungFuMaster* 中的四条想象轨迹。我们在四行中使用相同的条件帧（以绿色表示），并让世界模型想象其余部分。由于初始帧仅包含玩家，因此没有关于接下来会出现的敌人的信息。因此，世界模型会在每次模拟中生成不同类型和数量的对手。它还能反映一个重要的游戏机制（如蓝色框中所示），即第一个敌人在被玩家击中后会消失。

2 方法

我们将问题形式化为一个部分可观测马尔可夫决策过程（POMDP），包含图像观测

$x_t \in \mathbb{R}^{h \times w \times 3}$ 、离散动作 $a_t \in \{1, \dots, A\}$ 、标量奖励 $r_t \in \mathbb{R}$ 、情节终止 $d_t \in \{0, 1\}$ 、折扣因子 $\gamma \in (0, 1)$ 、初始观测分布 ρ_0 ，以及环境动力学 $x_{t+1}, r_t, d_t \sim p(x_{t+1}, r_t, d_t | x_{\leq t}, a_{\leq t})$ 。强化学习目标是训练一个策略 π ，使其产生的动作能最大化期望累积奖励 $\mathbb{E}_\pi[\sum_{t \geq 0} \gamma^t r_t]$ 。

我们的方法依赖于在想象中学习的三个标准组件 (Sutton & Barto, 2018): 经验收集、世界模型学习和行为学习。遵循 Ha & Schmidhuber (2018); Kaiser 等人 (2020); Hafner 等人 (2020; 2021) 的思路，我们的智能体学习仅在其世界模型内行动，并且我们仅利用真实经验来学习环境动力学。

我们反复执行以下三个步骤：

- 收集_经验：在真实环境中使用当前策略收集经验。
- 更新_世界_模型：改进奖励预测、回合终止判断以及下一观测值的预测。
- 更新_行为：在想象中改进策略和价值函数。

世界模型由一个离散自编码器 (Van Den Oord 等人, 2017) 构成，用于将图像转换为令牌并还原回图像，以及一个类GPT自回归Transformer (Vaswani 等人, 2017; Radford 等人, 2019; Brown 等人, 2020b)，其任务是捕捉环境动力学。图1展示了在想象过程中策略与这两个组件之间的交互。我们首先在2.1节和2.2节分别描述自编码器和Transformer，然后在第2.3节详细阐述在想象中学习策略和价值函数的具体过程。附录A提供了模型架构和超参数的全面描述。算法1总结了训练协议。

2.1 从图像观察到令牌

离散自编码器 (E, D) 学习了一种自己的符号语言，将高维图像表示为少量令牌。图1中用绿色箭头展示了帧与令牌之间的往返过程。

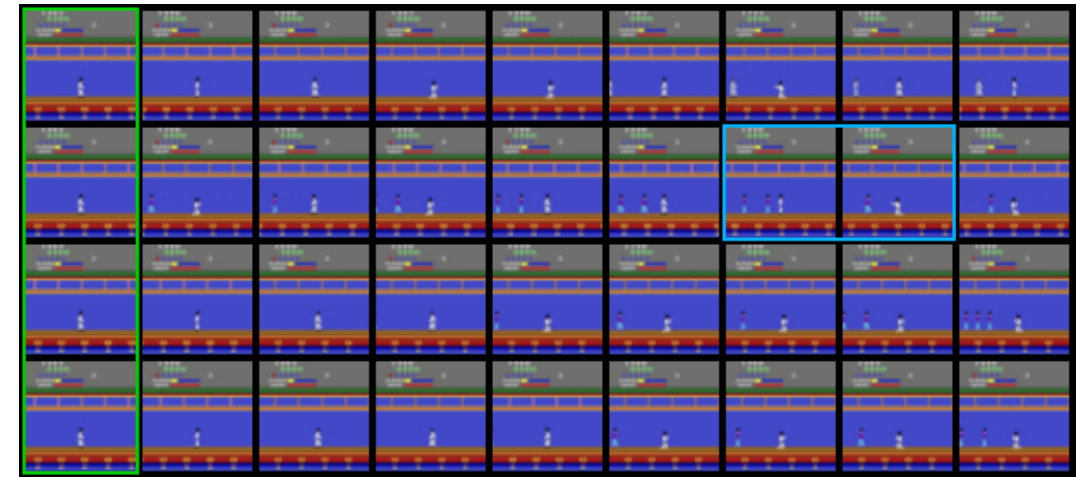


Figure 2: Four imagined trajectories in *KungFuMaster*. We use the same conditioning frame across the four rows, in green, and let the world model imagine the rest. As the initial frame only contains the player, there is no information about the enemies that will come next. Consequently, the world model generates different types and numbers of opponents in each simulation. It is also able to reflect an essential game mechanic, highlighted in the blue box, where the first enemy disappears after getting hit by the player.

2 METHOD

We formulate the problem as a Partially Observable Markov Decision Process (POMDP) with image observations $x_t \in \mathbb{R}^{h \times w \times 3}$, discrete actions $a_t \in \{1, \dots, A\}$, scalar rewards $r_t \in \mathbb{R}$, episode termination $d_t \in \{0, 1\}$, discount factor $\gamma \in (0, 1)$, initial observation distribution ρ_0 , and environment dynamics $x_{t+1}, r_t, d_t \sim p(x_{t+1}, r_t, d_t | x_{\leq t}, a_{\leq t})$. The reinforcement learning objective is to train a policy π that yields actions maximizing the expected sum of rewards $\mathbb{E}_\pi[\sum_{t \geq 0} \gamma^t r_t]$.

Our method relies on the three standard components to learn in imagination (Sutton & Barto, 2018): experience collection, world model learning, and behavior learning. In the vein of Ha & Schmidhuber (2018); Kaiser et al. (2020); Hafner et al. (2020; 2021), our agent learns to act exclusively within its world model, and we only make use of real experience to learn the environment dynamics.

We repeatedly perform the three following steps:

- collect_experience: gather experience in the real environment with the current policy.
- update_world_model: improve rewards, episode ends and next observations predictions.
- update_behavior: in imagination, improve the policy and value functions.

The world model is composed of a discrete autoencoder (Van Den Oord et al., 2017), to convert an image to tokens and back, and a GPT-like autoregressive Transformer (Vaswani et al., 2017; Radford et al., 2019; Brown et al., 2020b), whose task is to capture environment dynamics. Figure 1 illustrates the interplay between the policy and these two components during imagination. We first describe the autoencoder and the Transformer in Sections 2.1 and 2.2, respectively. Section 2.3 then details the procedure to learn the policy and value functions in imagination. Appendix A provides a comprehensive description of model architectures and hyperparameters. Algorithm 1 summarizes the training protocol.

2.1 FROM IMAGE OBSERVATIONS TO TOKENS

The discrete autoencoder (E, D) learns a symbolic language of its own to represent high-dimensional images as a small number of tokens. The back and forth between frames and tokens is illustrated with green arrows in Figure 1.

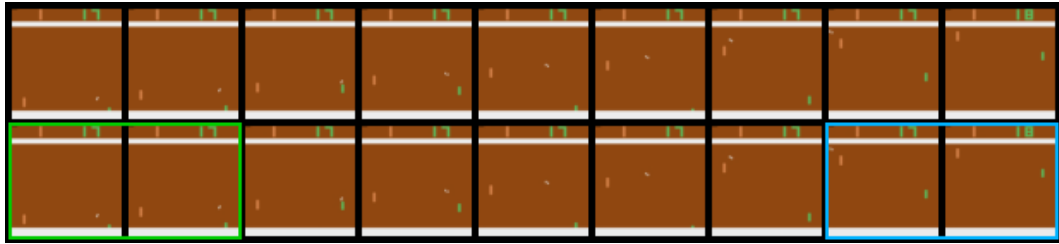


图3: *Pong*中的像素级完美预测。顶部行展示了在真实环境中收集的测试轨迹。底部行展示了该轨迹在世界模型中的重演。更具体地说，我们以真实序列的前两帧（绿色标记）作为世界模型的条件。然后，我们依次向其输入真实动作，让其想象后续帧。仅经过120场游戏的训练，世界模型就能完美模拟球的轨迹和玩家的移动。值得注意的是，它还捕捉到了赢得分后更新记分牌的游戏机制，如蓝色方框所示。

更准确地说，编码器 $E: \mathbb{R}^{h \times w \times 3} \rightarrow \{1, \dots, N\}^K$ 将输入图像 x_t 转换为来自大小为 N 的词汇表中的 K 个令牌。设 $\mathcal{E} = \{e_i\}_{i=1}^N \in \mathbb{R}^{N \times d}$ 为对应的 d 维向量嵌入表。输入图像 x_t 首先经过卷积神经网络（CNN）（LeCun等人，1989）处理，生成输出 $y_t \in \mathbb{R}^{K \times d}$ 。然后，我们获得输出令牌 $z_t = (z_t^1, \dots, z_t^K) \in \{1, \dots, N\}^K$ ，即 $z_t^k = \operatorname{argmin}_i \|y_t^k - e_i\|_2$ ，也就是 $\mathcal{E}$ 中最接近的嵌入向量的索引（Van Den Oord 等人，2017；Esser等人，2021）。相反，卷积神经网络解码器 $D: \{1, \dots, N\}^K \rightarrow \mathbb{R}^{h \times w \times 3}$ 将 K 个令牌重新转换回图像。

该离散自编码器在先前收集的帧上进行训练，使用由 L_1 重建损失、承诺损失（Van Den Oord等人，2017；Esser等人，2021）和感知损失（Esser等人，2021；Johnson等人，2016；Larsen等人，2016）组成的等权重组合。我们采用直通估计器（Bengio等人，2013）来实现反向传播训练。

2.2 建模动力学

从高层次来看，Transformer G 通过对离散自编码器随时间演变的语言进行建模，来捕捉环境动力学。其展开想象的核心作用在图1中用蓝色箭头突出显示。

具体而言， G 对交织的帧和动作令牌序列进行操作。输入序列 $(z_0^1, \dots, z_0^K, a_0, z_1^1, \dots, z_1^K, a_1, \dots, z_t^1, \dots, z_t^K, a_t)$ 是通过使用 E 对帧进行编码，从原始序列 $(x_0, a_0, x_1, a_1, \dots, x_t, a_t)$ 中获得的，如第2.1节所述。

在每个时间步 t ，Transformer 对以下三个分布进行建模：

$$\text{Transition: } \hat{z}_{t+1} \sim p_G(\hat{z}_{t+1} | z_{\leq t}, a_{\leq t}) \text{ with } \hat{z}_{t+1}^k \sim p_G(\hat{z}_{t+1}^k | z_{\leq t}, a_{\leq t}, z_{t+1}^{<k}) \quad (1)$$

$$\text{Reward: } \hat{r}_t \sim p_G(\hat{r}_t | z_{\leq t}, a_{\leq t}) \quad (2)$$

$$\text{Termination: } \hat{d}_t \sim p_G(\hat{d}_t | z_{\leq t}, a_{\leq t}) \quad (3)$$

请注意，对第 k 个令牌的条件设定还包括 $z_{t+1}^{<k} := (z_{t+1}^1, \dots, z_{t+1}^{k-1})$ ，即已经预测出的令牌，这意味着自回归过程发生在词元级别。

我们以自监督方式在来自过去经验的 L 时间步片段上训练 G 。对转移预测器和终止预测器使用交叉熵损失，对奖励预测器则根据奖励函数使用均方误差损失或交叉熵损失。

2.3 在想象中学习

离散自编码器 (E, D) 与 Transformer G 共同构成一个世界模型，具备想象能力。策略 π （在图1中以紫色箭头标示）专在这一想象MDP中进行学习。

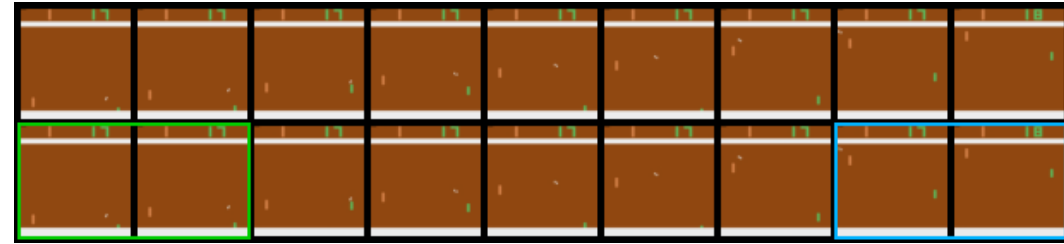


Figure 3: Pixel perfect predictions in *Pong*. The top row displays a test trajectory collected in the real environment. The bottom row depicts the reenactment of that trajectory inside the world model. More precisely, we condition the world model with the first two frames of the true sequence, in green. We then sequentially feed it the true actions and let it imagine the subsequent frames. After only 120 games of training, the world model perfectly simulates the ball’s trajectory and players’ movements. Notably, it also captures the game mechanic of updating the scoreboard after winning an exchange, as shown in the blue box.

More precisely, the encoder $E: \mathbb{R}^{h \times w \times 3} \rightarrow \{1, \dots, N\}^K$ converts an input image x_t into K tokens from a vocabulary of size N . Let $\mathcal{E} = \{e_i\}_{i=1}^N \in \mathbb{R}^{N \times d}$ be the corresponding embedding table of d -dimensional vectors. The input image x_t is first passed through a Convolutional Neural Network (CNN) (LeCun et al., 1989) producing output $y_t \in \mathbb{R}^{K \times d}$. We then obtain the output tokens $z_t = (z_t^1, \dots, z_t^K) \in \{1, \dots, N\}^K$ as $z_t^k = \operatorname{argmin}_i \|y_t^k - e_i\|_2$, the index of the closest embedding vector in $\mathcal{E}$ (Van Den Oord et al., 2017; Esser et al., 2021). Conversely, the CNN decoder $D: \{1, \dots, N\}^K \rightarrow \mathbb{R}^{h \times w \times 3}$ turns K tokens back into an image.

This discrete autoencoder is trained on previously collected frames, with an equally weighted combination of a L_1 reconstruction loss, a commitment loss (Van Den Oord et al., 2017; Esser et al., 2021), and a perceptual loss (Esser et al., 2021; Johnson et al., 2016; Larsen et al., 2016). We use a straight-through estimator (Bengio et al., 2013) to enable backpropagation training.

2.2 MODELING DYNAMICS

At a high level, the Transformer G captures the environment dynamics by modeling the language of the discrete autoencoder over time. Its central role of unfolding imagination is highlighted with the blue arrows in Figure 1.

Specifically, G operates over sequences of interleaved frame and action tokens. An input sequence $(z_0^1, \dots, z_0^K, a_0, z_1^1, \dots, z_1^K, a_1, \dots, z_t^1, \dots, z_t^K, a_t)$ is obtained from the raw sequence $(x_0, a_0, x_1, a_1, \dots, x_t, a_t)$ by encoding the frames with E , as described in Section 2.1.

At each time step t , the Transformer models the three following distributions:

$$\text{Transition: } \hat{z}_{t+1} \sim p_G(\hat{z}_{t+1} | z_{\leq t}, a_{\leq t}) \text{ with } \hat{z}_{t+1}^k \sim p_G(\hat{z}_{t+1}^k | z_{\leq t}, a_{\leq t}, z_{t+1}^{<k}) \quad (1)$$

$$\text{Reward: } \hat{r}_t \sim p_G(\hat{r}_t | z_{\leq t}, a_{\leq t}) \quad (2)$$

$$\text{Termination: } \hat{d}_t \sim p_G(\hat{d}_t | z_{\leq t}, a_{\leq t}) \quad (3)$$

Note that the conditioning for the k -th token also includes $z_{t+1}^{<k} := (z_{t+1}^1, \dots, z_{t+1}^{k-1})$, the tokens that were already predicted, i.e. the autoregressive process happens at the token level.

We train G in a self-supervised manner on segments of L time steps, sampled from past experience. We use a cross-entropy loss for the transition and termination predictors, and a mean-squared error loss or a cross-entropy loss for the reward predictor, depending on the reward function.

2.3 LEARNING IN IMAGINATION

Together, the discrete autoencoder (E, D) and the Transformer G form a world model, capable of imagination. The policy π , depicted with purple arrows in Figure 1, exclusively learns in this imagination MDP.

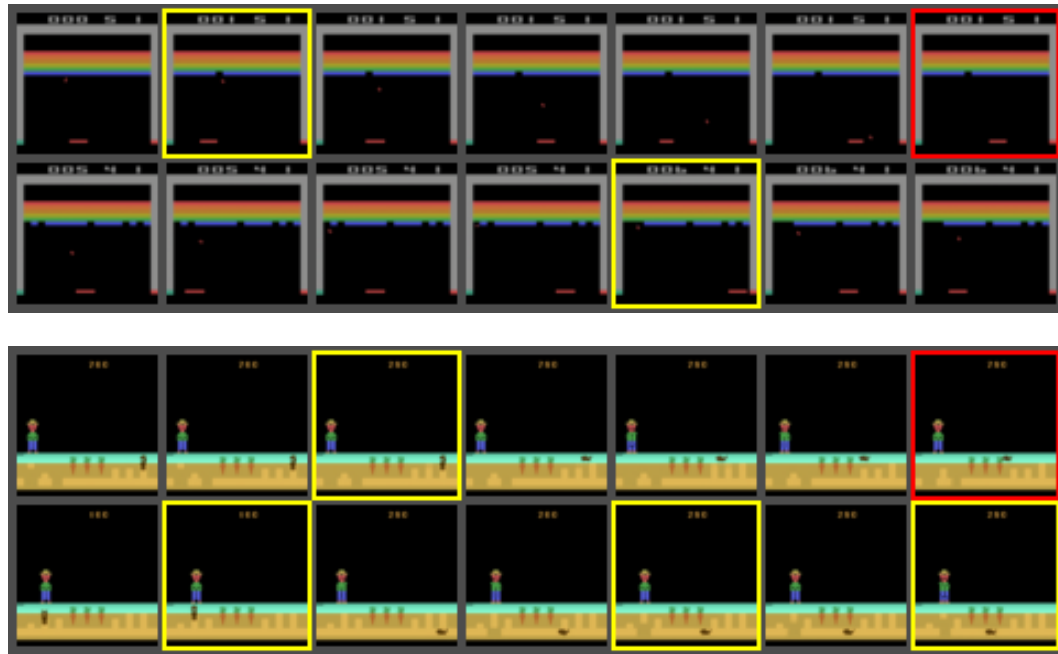


图4: 在*Breakout* (上)和*Gopher* (下)中想象奖励和情节终止。每一行展示了一条以真实环境中的单帧图像初始化的想象轨迹。黄色框表示世界模型预测正奖励的帧。在*Breakout*中, 模型捕捉到击碎砖块会带来奖励, 并且砖块在后续帧中被正确移除。在*Gopher*中, 玩家必须保护胡萝卜免受啮齿动物的侵害。世界模型成功内化了堵住洞口或杀死敌人会导致奖励这一规则。预测的回合终止用红色框标出。世界模型准确反映了在*Breakout*中未接住球, 或在*Gopher*中让敌人到达胡萝卜处, 都将导致一个回合的结束。

在时间步 t , 策略观测重建图像观测 $\hat{x}_t$ 并采样动作 $a_t \sim \pi(a_t | \hat{x}_{\leq t})$ 。世界模型随后预测奖励 $\hat{r}_t$ 、回合结束 d_t 以及下一观测 $\hat{x}_{t+1} = D(\hat{z}_{t+1})$, 其中 $\hat{z}_{t+1} \sim p_G(\hat{z}_{t+1} | z_0, a_0, \hat{z}_1, a_1, \dots, \hat{z}_t, a_t)$ 。这一想象过程从过去经验中采样的真实观测 x_0 开始初始化, 并展开 H 步, 即想象时间范围超参数。若在达到该时间范围前预测到回合结束, 则停止。图1展示了想象过程。

当我们在固定步数内展开想象时, 无法简单地使用蒙特卡洛估计来计算期望回报。因此, 为了对智能体在超过给定时间步后可能获得的奖励进行自举, 我们引入了一个价值网络 V , 用于估计 $V(\hat{x}_t) \simeq \mathbb{E}_\pi[\sum_{\tau \geq t} \gamma^{\tau-t} r_\tau]$ 。许多演员-评论家方法可用于在想象中训练 π 和 V (Sutton & Barto, 2018; Kaiser 等人, 2020; Hafner 等人, 2020)。为简便起见, 我们选择了DreamerV2 (Hafner 等人, 2021) 的学习目标和超参数, 该方法在Atari游戏中表现出色。附录B详细分解了强化学习目标。

3 实验

样本高效强化学习是一个不断发展的领域, 在复杂视觉环境中拥有多个基准 (Hafner, 2022; Kanervisto et al., 2022)。在本工作中, 我们专注于成熟的 Atari 100k基准 (Kaiser 等人, 2020)。我们在第3.1节中介绍了该基准及其基线。我们在第3.2节中描述了评估协议并讨论了结果。世界模型能力的定性示例在第3.3节中给出。

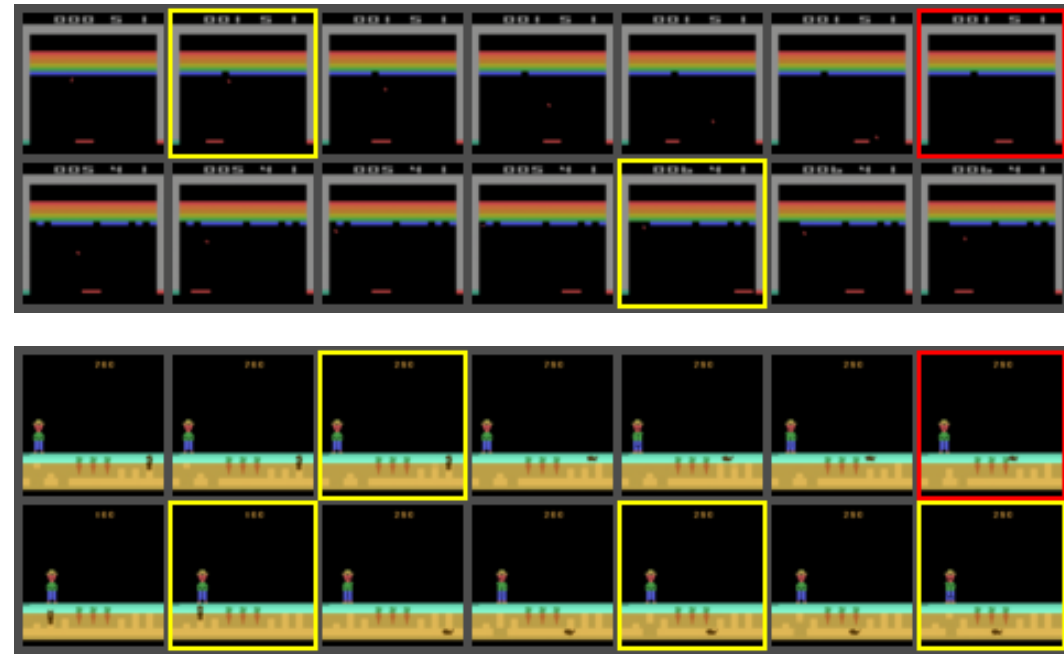


Figure 4: Imagining rewards and episode ends in *Breakout* (top) and *Gopher* (bottom). Each row depicts an imagined trajectory initialized with a single frame from the real environment. Yellow boxes indicate frames where the world model predicts a positive reward. In *Breakout*, it captures that breaking a brick yields rewards, and the brick is correctly removed from the following frames. In *Gopher*, the player has to protect the carrots from rodents. The world model successfully internalizes that plugging a hole or killing an enemy leads to rewards. Predicted episode terminations are highlighted with red boxes. The world model accurately reflects that missing the ball in *Breakout*, or letting an enemy reach the carrots in *Gopher*, will result in the end of an episode.

At time step t , the policy observes a reconstructed image observation $\hat{x}_t$ and samples action $a_t \sim \pi(a_t | \hat{x}_{\leq t})$. The world model then predicts the reward $\hat{r}_t$, the episode end d_t , and the next observation $\hat{x}_{t+1} = D(\hat{z}_{t+1})$, with $\hat{z}_{t+1} \sim p_G(\hat{z}_{t+1} | z_0, a_0, \hat{z}_1, a_1, \dots, \hat{z}_t, a_t)$. This imagination procedure is initialized with a real observation x_0 sampled from past experience, and is rolled out for H steps, the imagination horizon hyperparameter. We stop if an episode end is predicted before reaching the horizon. Figure 1 illustrates the imagination procedure.

As we roll out imagination for a fixed number of steps, we cannot simply use a Monte Carlo estimate for the expected return. Hence, to bootstrap the rewards that the agent would get beyond a given time step, we have a value network V that estimates $V(\hat{x}_t) \simeq \mathbb{E}_\pi[\sum_{\tau \geq t} \gamma^{\tau-t} \hat{r}_\tau]$.

Many actor-critic methods could be employed to train π and V in imagination (Sutton & Barto, 2018; Kaiser et al., 2020; Hafner et al., 2020). For the sake of simplicity, we opt for the learning objectives and hyperparameters of DreamerV2 (Hafner et al., 2021), that delivered strong performance in Atari games. Appendix B gives a detailed breakdown of the reinforcement learning objectives.

3 EXPERIMENTS

Sample-efficient reinforcement learning is a growing field with multiple benchmarks in complex visual environments (Hafner, 2022; Kanervisto et al., 2022). In this work, we focus on the well established Atari 100k benchmark (Kaiser et al., 2020). We present the benchmark and its baselines in Section 3.1. We describe the evaluation protocol and discuss the results in Section 3.2. Qualitative examples of the world model’s capabilities are given in Section 3.3.

表1: 在 Atari 100k 的 26个游戏中经过2小时实时经验后的回报, 以及人类标准化聚合指标。粗体数字表示不采用前瞻搜索的最优方法, 下划线数字则指定了整体最佳方法。IRIS 在超人水平游戏数量、平均值、四分位均值 (IQM) 和最优性差距方面均优于纯学习方法。

Game	前瞻搜索				无前瞻搜索				
	随机	人类	MuZero	EfficientZero	SimPLe	CURL	DrQ	SPR	IRIS (我们的)
Alien	227.8	7127.7	530.0	808.5	616.9	711.0	865.2	841.9	420.0
Amidar	5.8	1719.5	38.8	148.6	74.3	113.7	137.8	179.7	143.0
Assault	222.4	742.0	500.1	1263.1	527.2	500.9	579.6	565.6	1524.4
Asterix	210.0	8503.3	1734.0	<u>25557.8</u>	1128.3	567.2	763.6	962.5	853.6
BankHeist	14.2	753.1	192.5	351.0	34.2	65.3	232.9	345.4	53.1
BattleZone	2360.0	37187.5	7687.5	13871.2	4031.2	8997.8	10165.3	14834.1	13074.0
Boxing	0.1	12.1	15.1	52.7	7.8	0.9	9.0	35.7	70.1
Breakout	1.7	30.5	48.0	<u>414.1</u>	16.4	2.6	19.8	19.6	83.7
ChopperCommand	811.0	7387.8	1350.0	1117.3	979.4	783.5	844.6	946.3	1565.0
CrazyClimber	10780.5	35829.4	56937.0	<u>83940.2</u>	62583.6	9154.4	21539.0	36700.5	59324.2
DemonAttack	152.1	1971.0	3527.0	<u>13003.9</u>	208.1	646.5	1321.5	517.6	2034.4
Freeway	0.0	29.6	21.8	21.8	16.7	28.3	20.3	19.3	31.1
Frostbite	65.2	4334.7	255.0	296.3	236.9	1226.5	1014.2	1170.7	259.1
Gopher	257.6	2412.5	1256.0	<u>3260.3</u>	596.8	400.9	621.6	660.6	2236.1
Hero	1027.0	30826.4	3095.0	<u>9315.9</u>	2656.6	4987.7	4167.9	5858.6	7037.4
Jamesbond	29.0	302.8	87.5	<u>517.0</u>	100.5	331.0	349.1	366.5	462.7
Kangaroo	52.0	3035.0	62.5	724.1	51.2	740.2	1088.4	3617.4	838.2
Krull	1598.0	2665.5	4890.8	5663.3	2204.8	3049.2	4402.1	3681.6	6616.4
KungFuMaster	258.5	22736.3	18813.0	<u>30944.8</u>	14862.5	8155.6	11467.4	14783.2	21759.8
MsPacman	307.3	6951.6	1265.6	1281.2	1480.0	1064.0	1218.1	1318.4	999.1
Pong	-20.7	14.6	-6.7	<u>20.1</u>	12.8	-18.5	-9.1	-5.4	14.6
PrivateEye	24.9	69571.3	56.3	96.7	35.0	81.9	3.5	86.0	100.0
Qbert	163.9	13455.0	3952.0	<u>13781.9</u>	1288.8	727.0	1810.7	866.3	745.7
RoadRunner	11.5	7845.0	2500.0	<u>17751.3</u>	5640.6	5006.1	11211.4	12213.1	9614.6
Seaquest	68.4	42054.7	208.0	<u>1100.2</u>	683.3	315.2	352.3	558.1	661.3
UpNDown	533.4	11693.2	2896.9	<u>17264.2</u>	3350.3	2646.4	4324.5	10859.2	3546.2
#超人人类 (↑)	0	N/A	5	14	1	2	3	6	10
平均值 (↑)	0.000	1.000	0.562	<u>1.943</u>	0.332	0.261	0.465	0.616	1.046
中位数 (↑)	0.000	1.000	0.227	<u>1.090</u>	0.134	0.092	0.313	0.396	0.289
IQM (↑)	0.000	1.000	N/A	N/A	0.130	0.113	0.280	0.337	0.501
最优性差距 (↓)	1.000	0.000	N/A	N/A	0.729	0.768	0.631	0.577	0.512

3.1 基准与基线

Atari 100k基准包含26款具有不同机制的Atari游戏 (Bellemare 等人, 2013a), 用于评估智能体广泛的能力范围。在该基准中, 智能体在每个环境中仅被允许执行10万次动作。这一限制大致相当于人类2小时的游戏时间。相比之下, 无限制的Atari智能体通常需要训练5000万步, 经验量增加了500倍。

在Atari 100k基准上比较了多个基线方法。SimPLe (Kaiser 等人, 2020) 在视频生成模型中通过PPO (Schulman 等人, 2017) 训练策略。CURL (Laskin 等人, 2020) 利用对比学习从高级图像特征中开发离策略智能体。DrQ (Yarats 等人, 2021) 对输入图像进行数据增强, 并在多次变换上平均Q值估计。SPR (Schwarzer等人, 2021) 在增强视图和相邻时间步上强制执行输入图像的一致性表征。上述基线方法还采用了其他技术来提升性能, 例如优先经验回放 (Schaul 等人, 2016)、 ϵ -贪心调度或数据增强。

我们对有无前瞻搜索的方法进行了区分。确实, 依赖决策时进行搜索的算法 (Silver等人, 2016; 2018; Schrittwieser等人, 2020) 能够大幅提升智能体性能, 但它们在计算资源和代码复杂度方面也付出了高昂代价。MuZero (Schrittwieser等人, 2020) 和EfficientZero (Ye等人, 2021) 是目前Atari 100k中基于搜索方法的标准。MuZero利用蒙特卡洛树搜索 (MCTS) (Kocsis & Szepesvári, 2006; Coulom, 2007) 作为策略改进算子, 通过在世界模型的潜在空间中展开多条假设轨迹来实现。EfficientZero在MuZero基础上进行了改进, 引入了自监督一致性损失, 能够一步预测短时域内的回报, 并通过其世界模型修正离线策略轨迹。

Table 1: Returns on the 26 games of Atari 100k after 2 hours of real-time experience, and human-normalized aggregate metrics. Bold numbers indicate the top methods without lookahead search while underlined numbers specify the overall best methods. IRIS outperforms learning-only methods in terms of number of superhuman games, mean, interquartile mean (IQM), and optimality gap.

Game	Lookahead search				No lookahead search				
	Random	Human	MuZero	EfficientZero	SimPLe	CURL	DrQ	SPR	IRIS (ours)
Alien	227.8	7127.7	530.0	808.5	616.9	711.0	865.2	841.9	420.0
Amidar	5.8	1719.5	38.8	148.6	74.3	113.7	137.8	179.7	143.0
Assault	222.4	742.0	500.1	1263.1	527.2	500.9	579.6	565.6	1524.4
Asterix	210.0	8503.3	1734.0	<u>25557.8</u>	1128.3	567.2	763.6	962.5	853.6
BankHeist	14.2	753.1	192.5	351.0	34.2	65.3	232.9	345.4	53.1
BattleZone	2360.0	37187.5	7687.5	13871.2	4031.2	8997.8	10165.3	14834.1	13074.0
Boxing	0.1	12.1	15.1	52.7	7.8	0.9	9.0	35.7	70.1
Breakout	1.7	30.5	48.0	<u>414.1</u>	16.4	2.6	19.8	19.6	83.7
ChopperCommand	811.0	7387.8	1350.0	1117.3	979.4	783.5	844.6	946.3	1565.0
CrazyClimber	10780.5	35829.4	56937.0	<u>83940.2</u>	62583.6	9154.4	21539.0	36700.5	59324.2
DemonAttack	152.1	1971.0	3527.0	<u>13003.9</u>	208.1	646.5	1321.5	517.6	2034.4
Freeway	0.0	29.6	21.8	21.8	16.7	28.3	20.3	19.3	31.1
Frostbite	65.2	4334.7	255.0	296.3	236.9	1226.5	1014.2	1170.7	259.1
Gopher	257.6	2412.5	1256.0	<u>3260.3</u>	596.8	400.9	621.6	660.6	2236.1
Hero	1027.0	30826.4	3095.0	<u>9315.9</u>	2656.6	4987.7	4167.9	5858.6	7037.4
Jamesbond	29.0	302.8	87.5	<u>517.0</u>	100.5	331.0	349.1	366.5	462.7
Kangaroo	52.0	3035.0	62.5	724.1	51.2	740.2	1088.4	3617.4	838.2
Krull	1598.0	2665.5	4890.8	5663.3	2204.8	3049.2	4402.1	3681.6	6616.4
KungFuMaster	258.5	22736.3	18813.0	<u>30944.8</u>	14862.5	8155.6	11467.4	14783.2	21759.8
MsPacman	307.3	6951.6	1265.6	1281.2	1480.0	1064.0	1218.1	1318.4	999.1
Pong	-20.7	14.6	-6.7	<u>20.1</u>	12.8	-18.5	-9.1	-5.4	14.6
PrivateEye	24.9	69571.3	56.3	96.7	35.0	81.9	3.5	86.0	100.0
Qbert	163.9	13455.0	3952.0	<u>13781.9</u>	1288.8	727.0	1810.7	866.3	745.7
RoadRunner	11.5	7845.0	2500.0	<u>17751.3</u>	5640.6	5006.1	11211.4	12213.1	9614.6
Seaquest	68.4	42054.7	208.0	<u>1100.2</u>	683.3	315.2	352.3	558.1	661.3
UpNDown	533.4	11693.2	2896.9	<u>17264.2</u>	3350.3	2646.4	4324.5	10859.2	3546.2
#Superhuman (↑)	0	N/A	5	14	1	2	3	6	10
Mean (↑)	0.000	1.000	0.562	<u>1.943</u>	0.332	0.261	0.465	0.616	1.046
Median (↑)	0.000	1.000	0.227	<u>1.090</u>	0.134	0.092	0.313	0.396	0.289
IQM (↑)	0.000	1.000	N/A	N/A	0.130	0.113	0.280	0.337	0.501
Optimality Gap (↓)	1.000	0.000	N/A	N/A	0.729	0.768	0.631	0.577	0.512

3.1 BENCHMARK AND BASELINES

Atari 100k consists of 26 Atari games (Bellemare et al., 2013a) with various mechanics, evaluating a wide range of agent capabilities. In this benchmark, an agent is only allowed 100k actions in each environment. This constraint is roughly equivalent to 2 hours of human gameplay. By way of comparison, unconstrained Atari agents are usually trained for 50 million steps, a 500 fold increase in experience.

Multiple baselines were compared on the Atari 100k benchmark. SimPLe (Kaiser et al., 2020) trains a policy with PPO (Schulman et al., 2017) in a video generation model. CURL (Laskin et al., 2020) develops off-policy agents from high-level image features obtained with contrastive learning. DrQ (Yarats et al., 2021) augments input images and averages Q-value estimates over several transformations. SPR (Schwarzer et al., 2021) enforces consistent representations of input images across augmented views and neighbouring time steps. The aforementioned baselines carry additional techniques to improve performance, such as prioritized experience replay (Schaul et al., 2016), epsilon-greedy scheduling, or data augmentation.

We make a distinction between methods with and without lookahead search. Indeed, algorithms relying on search at decision time (Silver et al., 2016; 2018; Schrittwieser et al., 2020) can vastly improve agent performance, but they come at a premium in computational resources and code complexity. MuZero (Schrittwieser et al., 2020) and EfficientZero (Ye et al., 2021) are the current standard for search-based methods in Atari 100k. MuZero leverages Monte Carlo Tree Search (MCTS) (Kocsis & Szepesvári, 2006; Coulom, 2007) as a policy improvement operator, by unrolling multiple hypothetical trajectories in the latent space of a world model. EfficientZero improves upon MuZero by introducing a self-supervised consistency loss, predicting returns over short horizons in one shot, and correcting off-policy trajectories with its world model.

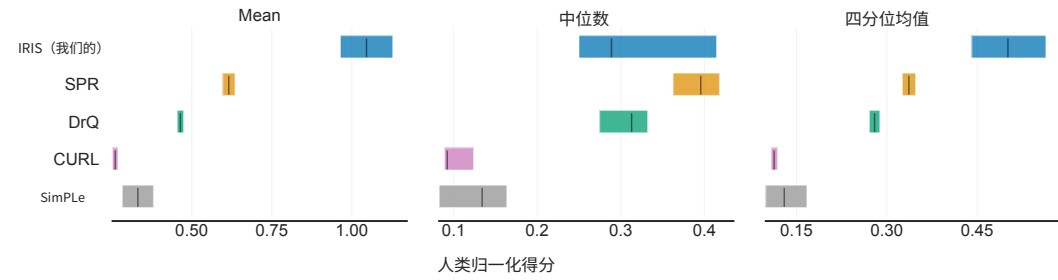
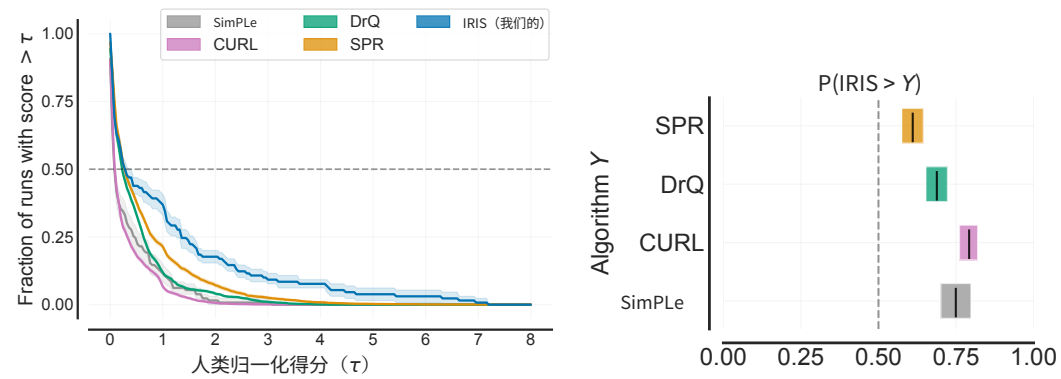


图 5: 平均值、中位数和四分位均值的人类标准化分数, 使用分层自助法置信区间计算。IRIS 和 SimPLe 各运行 5 次, SPR、CURL 和 DrQ 各运行 100 次 (Agarwal 等人, 2021)。



(a) 性能剖面图, 即超过给定人类归一化得分的运行比例。 (b) 改进概率, 即 IRIS 在任何游戏中超越基线的可能性。人类归一化得分。 是让 IRIS 在任何游戏上超越基线。

图6: 性能剖面图 (左) 和改进概率 (右) (Agarwal 等人, 2021)。

3.2 结果

人类归一化得分是 Atari 100k 中既定的性能衡量指标。它定义为 $\frac{score_agent - score_random}{score_human - score_random}$, 其中分数_随机 来自随机策略, 而分数_人类 是从人类玩家 (Wang 等人, 2016) 那里获得的。

表 1 展示了各游戏的回报以及人类归一化聚合指标。对于 MuZero 和 EfficientZero, 我们报告了 Ye 等人 (2021) 发表的平均结果 (3 次运行)。对于其他基线 (CURL、DrQ、SPR 为 100 次新运行, SimPLe 为 5 次已有运行), 我们使用了 Agarwal 等人 (2021) 在 Atari 100k 案例研究中取得的结果。最后, 我们通过对每个游戏 (5 次运行) 在训练结束时收集的 100 个回合进行平均, 来评估 IRIS。

Agarwal 等人 (2021) 讨论了平均值和中位数分数的局限性, 并指出在强化学习基准中, 标准点估计和区间估计之间存在显著差异。遵循其建议, 我们在图 5 中总结了人类标准化分数, 并附带了针对平均值、中位数和四分位均值 (IQM) 的分层自助法置信区间。为了进行更精细的比较, 我们还在图 6 中提供了性能剖面图和改进概率。

仅需相当于两小时的游戏时间, IRIS 就取得了 1.046 的超人类平均得分 (+70%), IQM 为 0.501 (+49%), 最优性差距为 0.512 (+11%), 并在 26 款游戏中以 10 款优于人类玩家 (+67%), 其中相对改进程度是相对于 SPR (Schwarzer 等人, 2021) 计算的。这些结果构成了 Atari 100k 基准中无前瞻搜索方法的最先进水平。我们还注意到, IRIS 优于 MuZero, 尽管后者并非针对样本高效区域而设计。

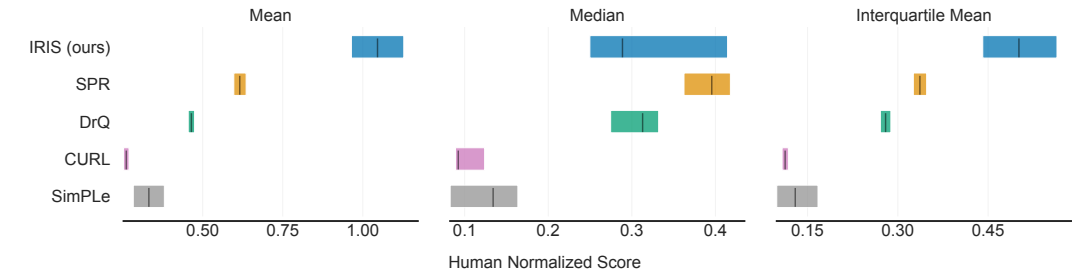
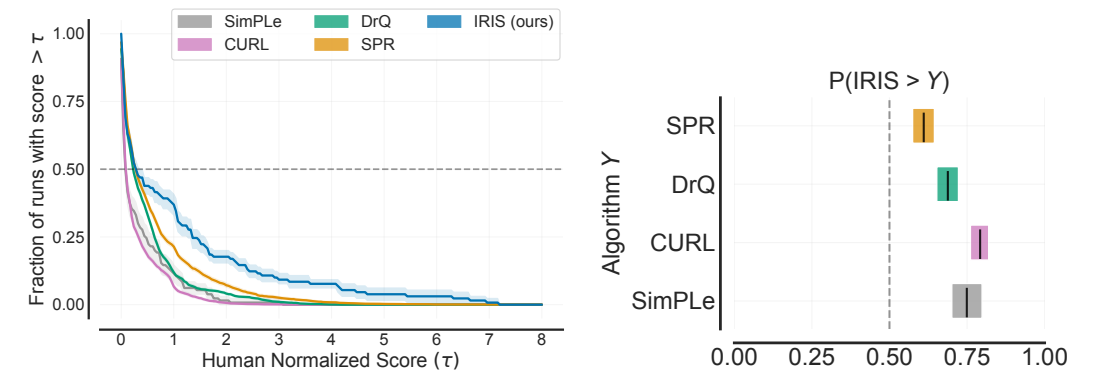


Figure 5: Mean, median, and interquartile mean human normalized scores, computed with stratified bootstrap confidence intervals. 5 runs for IRIS and SimPLe, 100 runs for SPR, CURL, and DrQ (Agarwal et al., 2021).



(a) Performance profiles, i.e. fraction of runs above a given human normalized score. (b) Probabilities of improvement, i.e. how likely it is for IRIS to outperform baselines on any game.

Figure 6: Performance profiles (left) and probabilities of improvement (right) (Agarwal et al., 2021).

3.2 RESULTS

The human normalized score is the established measure of performance in Atari 100k. It is defined as $\frac{score_agent - score_random}{score_human - score_random}$, where $score_random$ comes from a random policy, and $score_human$ is obtained from human players (Wang et al., 2016).

Table 1 displays returns across games and human-normalized aggregate metrics. For MuZero and EfficientZero, we report the averaged results published by Ye et al. (2021) (3 runs). We use results from the Atari 100k case study conducted by Agarwal et al. (2021) for the other baselines (100 new runs for CURL, DrQ, SPR, and 5 existing runs for SimPLe). Finally, we evaluate IRIS by computing an average over 100 episodes collected at the end of training for each game (5 runs).

Agarwal et al. (2021) discuss the limitations of mean and median scores, and show that substantial discrepancies arise between standard point estimates and interval estimates in RL benchmarks. Following their recommendations, we summarize in Figure 5 the human normalized scores with stratified bootstrap confidence intervals for mean, median, and interquartile mean (IQM). For finer comparisons, we also provide performance profiles and probabilities of improvement in Figure 6.

With the equivalent of only two hours of gameplay, IRIS achieves a superhuman mean score of 1.046 (+70%), an IQM of 0.501 (+49%), an optimality gap of 0.512 (+11%), and outperforms human players on 10 out of 26 games (+67%), where the relative improvements are computed with respect to SPR (Schwarzer et al., 2021). These results constitute a new state of the art for methods without lookahead search in the Atari 100k benchmark. We also note that IRIS outperforms MuZero, although the latter was not designed for the sample-efficient regime.

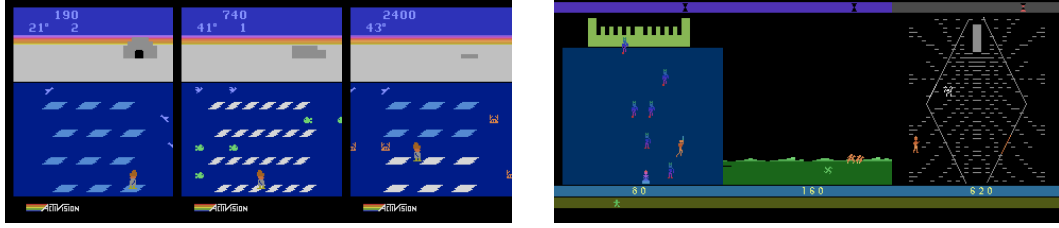


图7: 游戏*Frostbite* (左) 和*Krull* (右) 中三个连续关卡。在实验中, 世界模型难以模拟*Frostbite*后续关卡, 但在*Krull*中则无此问题。实际上, 要离开*Frostbite* 第一关, 需要执行一长串且概率极低的动作序列: 先建造冰屋, 再从屏幕底部返回冰屋。此类罕见事件阻碍了世界模型内化游戏的新特征, 导致策略在想象中无法体验这些内容。虽然*Krull* 拥有更多样化的关卡, 但世界模型成功反映了这种多样性, IRIS甚至在此环境中树立了新的最先进水平。这很可能是因为, 在*Krull*中, 从一个阶段到下一阶段的转换更为频繁, 从而充分覆盖了每个关卡。

此外, 性能剖面图 (图 6a) 显示, IRIS 在其表现最差的 50% 游戏中与最强基线持平, 并且在此之后随机占据优势 (Agarwal 等人, 2021; Dror 等人, 2019) 优于其他方法。类似地, 对于所有基线而言, 改进的概率均大于 0.5 (图 6b)。

就中位数分数而言, IRIS与其他方法存在重叠 (图5)。有趣的是, Schwarzer等人 (2021) 指出, 中位数仅受少数决定性游戏影响, 这一点从DrQ、CURL和SPR即使在100次运行中, 其中位数分数的置信区间宽度也可以看出。

我们观察到, IRIS 在那些随着训练进行不会出现分布偏移的游戏中表现尤为出色。此类游戏的示例包括 *Pong*、*Breakout* 和 *Boxing*。相反, 当通过一个低概率事件解锁新关卡或新游戏机制时, 智能体就会陷入困境。这揭示了一个双重探索问题。IRIS 必须首先发现游戏的某个新方面, 以便其世界模型将其内化。只有这样, 策略才有可能重新发现并加以利用。图 7 在《*Frostbite*》和《*Krull*》这两个包含多个关卡的游戏中详细描述了这一现象。总之, 只要关卡之间的过渡不依赖于低概率事件, 双重探索问题就不会阻碍性能表现。

另一类难以模拟的游戏是视觉上具有挑战性的环境, 在这些环境中捕捉微小细节至关重要。如附录E所述, 增加用于编码帧的令牌数量会提升性能, 但代价是计算量增加。

3.3 世界模型分析

由于 IRIS 完全在想象中学习行为, 世界模型的质量是我们方法的基石。例如, 离散自编码器能否正确重构像球、玩家或敌人这样的元素至关重要。同样, Transformer 无法捕捉重要的游戏机制 (如奖励归因或情节终止) 可能会严重削弱智能体的性能。因此, 无论想象轨迹的数量有多少, 如果世界模型存在缺陷, 智能体最终学到的策略也将是次优的。

尽管第3.2节提供了定量评估, 但我们旨在通过世界模型能力的定性示例来补充分析。图2展示了在不确定性下生成多种可能未来的情形。图3描绘了*Pong*中的像素完美预测。最后, 我们在图4中展示了对奖励和回合终止的预测, 这些预测对强化学习目标至关重要。

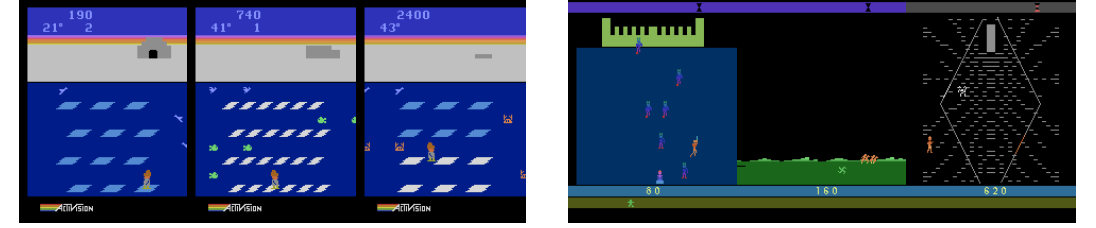


Figure 7: Three consecutive levels in the games *Frostbite* (left) and *Krull* (right). In our experiments, the world model struggles to simulate subsequent levels in *Frostbite*, but not in *Krull*. Indeed, exiting the first level in *Frostbite* requires a long and unlikely sequence of actions to first build the igloo, and then go back to it from the bottom of the screen. Such rare events prevent the world model from internalizing new aspects of the game, which will therefore not be experienced by the policy in imagination. While *Krull* features more diverse levels, the world model successfully reflects this variety, and IRIS even sets a new state of the art in this environment. This is likely due to more frequent transitions from one stage to the next in *Krull*, resulting in a sufficient coverage of each level.

In addition, performance profiles (Figure 6a) reveal that IRIS is on par with the strongest baselines for its bottom 50% of games, at which point it stochastically dominates (Agarwal et al., 2021; Dror et al., 2019) the other methods. Similarly, the probability of improvement is greater than 0.5 for all baselines (Figure 6b).

In terms of median score, IRIS overlaps with other methods (Figure 5). Interestingly, Schwarzer et al. (2021) note that the median is only influenced by a few decisive games, as evidenced by the width of the confidence intervals for median scores, even with 100 runs for DrQ, CURL and SPR.

We observe that IRIS is particularly strong in games that do not suffer from distributional shifts as the training progresses. Examples of such games include *Pong*, *Breakout*, and *Boxing*. On the contrary, the agent struggles when a new level or game mechanic is unlocked through an unlikely event. This sheds light on a double exploration problem. IRIS has to first discover a new aspect of the game for its world model to internalize it. Only then may the policy rediscover and exploit it. Figure 7 details this phenomenon in *Frostbite* and *Krull*, two games with multiple levels. In summary, as long as transitions between levels do not depend on low-probability events, the double exploration problem does not hinder performance.

Another kind of games difficult to simulate are visually challenging environments where capturing small details is important. As discussed in Appendix E, increasing the number of tokens to encode frames improves performance, albeit at the cost of increased computation.

3.3 WORLD MODEL ANALYSIS

As IRIS learns behaviors entirely in its imagination, the quality of the world model is the cornerstone of our approach. For instance, it is key that the discrete autoencoder correctly reconstructs elements like a ball, a player, or an enemy. Similarly, the potential inability of the Transformer to capture important game mechanics, like reward attribution or episode termination, can severely hamper the agent’s performance. Hence, no matter the amount of imagined trajectories, the agent will learn suboptimal policies if the world model is flawed.

While Section 3.2 provides a quantitative evaluation, we aim to complement the analysis with qualitative examples of the abilities of the world model. Figure 2 shows the generation of many plausible futures in the face of uncertainty. Figure 3 depicts pixel-perfect predictions in *Pong*. Finally, we illustrate in Figure 4 predictions for rewards and episode terminations, which are crucial to the reinforcement learning objective.

4 相关工作

在世界模型想象中的学习

在学习的世界模型中训练策略的想法最初是在表格环境中进行研究的(Sutton & Barto, 2018)。Ha & Schmidhuber (2018) 表明, 可以通过自编码器和循环网络模拟简单的视觉环境。SimPLe (Kaiser 等人, 2020) 证明, 在视频预测模型中训练的PPO策略(Schulman 等人, 2017) 在某些Atari游戏上超越了人类。在Dreamer (Hafner 等人, 2020) 的基础上, DreamerV2 (Hafner 等人, 2021) 成为首个通过想象学习达到Atari 50M基准人类水平表现的智能体。其世界模型将卷积自编码器与用于潜在动力学学习的循环状态空间模型 (RSSM) (Hafner 等人, 2019) 相结合。最近, Chen 等人 (2022) 探索了DreamerV2的一种变体, 其中Transformer取代了RSSM中的循环网络; 而Seo等人 (2022) 则在存在可用于预训练的离线视频数据集的情况下增强了DreamerV2。

基于Transformer的强化学习

继自然语言处理领域取得显著进展之后 (Manning & Goldie, 2022), 强化学习社区近期也踏入了Transformer的领域。Parisotto等人 (2020) 发现, 标准的Transformer架构难以通过强化学习目标进行优化。作者提出用门控层替代残差连接以稳定学习过程。我们的世界模型无需此类修改, 这很可能归功于其自监督学习目标。轨迹Transformer (Janner等人, 2021) 和决策Transformer (Chen等人, 2021) 将离线轨迹表示为静态序列数据集, 而在线决策Transformer (Zheng等人, 2022) 则将其扩展到在线场景。轨迹Transformer经过训练可预测未来回报、状态和动作。在推理时, 它能够通过奖励驱动的束搜索规划最优动作, 但该方法仅限于低维状态空间。相比之下, 决策Transformer可以处理图像输入, 但难以作为世界模型进行扩展。Ozair等人 (2021) 引入了MuZero (Schrittwieser等人, 2020) 的离线变体, 通过在Transformer上对动作和轨迹级离散潜在变量进行混合搜索, 具备了处理随机环境的能力。

基于离散自编码器和Transformer的视频生成

VQGAN (Esser等人, 2021) 和DALL-E (Ramesh等人, 2021) 使用离散自编码器将帧压缩为短序列的令牌, 然后Transformer可以自回归地对其进行建模。其他工作将这种方法扩展到视频生成。GODIVA (Wu等人, 2021) 针对文本条件视频生成, 对帧序列而非单帧进行建模。VideoGPT (Yan等人, 2021) 引入了视频级离散自编码器以及具有空间和时间注意力模式的Transformer, 用于无条件与动作条件视频生成。

5 结论

我们提出了IRIS, 这是一种在由离散自编码器和自回归Transformer组成的世界模型的想象中进行纯学习的智能体。IRIS在Atari 100k基准上为不使用前瞻搜索的方法树立了新的最先进水平。我们展示了其世界模型对游戏机制具有深刻理解, 从而在某些游戏中实现了像素级完美预测。我们还展示了世界模型的生成能力, 当在想象中进行训练时, 提供了丰富的游戏体验。最终, 与现有久经考验的智能体相比, IRIS只需极少的调优, 为高效解决复杂环境开辟了新路径。

未来, IRIS 可以扩展到需要从世界模型的速度中受益的计算密集型和具有挑战性的任务。此外, 其策略目前从重构帧中学习, 但或许可以利用世界模型的内部表示。另一个令人兴奋的研究方向是将想象中学习与蒙特卡洛树搜索 (MCTS) 相结合。事实上, 两种方法都取得了令人印象深刻的结果, 且它们对智能体性能的贡献可能是互补的。

4 RELATED WORK

LEARNING IN THE IMAGINATION OF WORLD MODELS

The idea of training policies in a learnt model of the world was first investigated in tabular environments (Sutton & Barto, 2018). Ha & Schmidhuber (2018) showed that simple visual environments could be simulated with autoencoders and recurrent networks. SimPLe (Kaiser et al., 2020) demonstrated that a PPO policy (Schulman et al., 2017) trained in a video prediction model outperformed humans in some Atari games. Improving upon Dreamer (Hafner et al., 2020), DreamerV2 (Hafner et al., 2021) was the first agent learning in imagination to achieve human-level performance in the Atari 50M benchmark. Its world model combines a convolutional autoencoder with a recurrent state-space model (RSSM) (Hafner et al., 2019) for latent dynamics learning. More recently, Chen et al. (2022) explored a variant of DreamerV2 where a Transformer replaces the recurrent network in the RSSM and Seo et al. (2022) enhance DreamerV2 in the setting where an offline dataset of videos is available for pretraining.

REINFORCEMENT LEARNING WITH TRANSFORMERS

Following spectacular advances in natural language processing (Manning & Goldie, 2022), the reinforcement learning community has recently stepped into the realm of Transformers. Parisotto et al. (2020) make the observation that the standard Transformer architecture is difficult to optimize with RL objectives. The authors propose to replace residual connections by gating layers to stabilize the learning procedure. Our world model does not require such modifications, which is most likely due to its self-supervised learning objective. The Trajectory Transformer (Janner et al., 2021) and the Decision Transformer (Chen et al., 2021) represent offline trajectories as a static dataset of sequences, and the Online Decision Transformer (Zheng et al., 2022) extends the latter to the online setting. The Trajectory Transformer is trained to predict future returns, states and actions. At inference time, it can thus plan for the optimal action with a reward-driven beam search, yet the approach is limited to low-dimensional states. On the contrary, Decision Transformers can handle image inputs but cannot be easily extended as world models. Ozair et al. (2021) introduce an offline variant of MuZero (Schrittwieser et al., 2020) capable of handling stochastic environments by performing an hybrid search with a Transformer over both actions and trajectory-level discrete latent variables.

VIDEO GENERATION WITH DISCRETE AUTOENCODERS AND TRANSFORMERS

VQGAN (Esser et al., 2021) and DALL-E (Ramesh et al., 2021) use discrete autoencoders to compress a frame into a small sequence of tokens, that a transformer can then model autoregressively. Other works extend the approach to video generation. GODIVA (Wu et al., 2021) models sequences of frames instead of a single frame for text conditional video generation. VideoGPT (Yan et al., 2021) introduces video-level discrete autoencoders, and Transformers with spatial and temporal attention patterns, for unconditional and action conditional video generation.

5 CONCLUSION

We introduced IRIS, an agent that learns purely in the imagination of a world model composed of a discrete autoencoder and an autoregressive Transformer. IRIS sets a new state of the art in the Atari 100k benchmark for methods without lookahead search. We showed that its world model acquires a deep understanding of game mechanics, resulting in pixel perfect predictions in some games. We also illustrated the generative capabilities of the world model, providing a rich gameplay experience when training in imagination. Ultimately, with minimal tuning compared to existing battle-hardened agents, IRIS opens a new path towards efficiently solving complex environments.

In the future, IRIS could be scaled up to computationally demanding and challenging tasks that would benefit from the speed of its world model. Besides, its policy currently learns from reconstructed frames, but it could probably leverage the internal representations of the world model. Another exciting avenue of research would be to combine learning in imagination with MCTS. Indeed, both approaches deliver impressive results, and their contributions to agent performance might be complementary.

可重复性声明

不同的组件及其训练目标在第2节和附录B中介绍。我们在附录A中描述了模型架构并列出了超参数。我们在附录G中指定了用于生成结果所需的资源。算法1明确了训练循环中各组件之间的相互作用。在第3.2节中，我们提供了基线报告结果的来源以及评估协议。

代码已开源，以确保结果可重复并促进未来研究。运行代码库所需依赖极少，我们提供了详尽的用户指南帮助入门。训练和评估可通过简单命令启动，支持通过配置文件进行自定义，并包含可视化智能体运行过程的脚本，允许用户与世界模型交互。

伦理声明

针对真实环境开发自主智能体引发了许多安全与环境问题。在训练期间，智能体可能对个体造成严重伤害，并破坏其周围环境。我们认为，在世界模型想象中的学习能够极大降低训练新型自主智能体相关的风险。事实上，在本研究中，我们提出了一种能够以极少量样本准确建模环境的世界模型架构。然而，在未来的研究方向上，可以更进一步利用现有数据，从而消除与现实世界交互的必要性。

致谢

我们感谢Maxim Peter、Bálint Máté、Daniele Paliotta、Atul Sinha和Alexandre Dupuis提供的深刻讨论与宝贵意见。Vincent Micheli获得了瑞士国家科学基金会基金编号FNS-187494的资助。

REPRODUCIBILITY STATEMENT

The different components and their training objectives are introduced in Section 2 and Appendix B. We describe model architectures and list hyperparameters in Appendix A. We specify the resources used to produce our results in Appendix G. Algorithm 1 makes explicit the interplay between components in the training loop. In Section 3.2, we provide the source of the reported results for the baselines, as well as the evaluation protocol.

The code is open-source to ensure reproducible results and foster future research. Minimal dependencies are required to run the codebase and we provide a thorough user guide to get started. Training and evaluation can be launched with simple commands, customization is possible with configuration files, and we include scripts to visualize agents playing and let users interact with the world model.

ETHICS STATEMENT

The development of autonomous agents for real-world environments raises many safety and environmental concerns. During its training period, an agent may cause serious harm to individuals and damage its surroundings. It is our belief that learning in the imagination of world models greatly reduces the risks associated with training new autonomous agents. Indeed, in this work, we propose a world model architecture capable of accurately modeling environments with very few samples. However, in a future line of research, one could go one step further and leverage existing data to eliminate the necessity of interacting with the real world.

ACKNOWLEDGMENTS

We would like to thank Maxim Peter, Bálint Máté, Daniele Paliotta, Atul Sinha, and Alexandre Dupuis for insightful discussions and comments. Vincent Micheli was supported by the Swiss National Science Foundation under grant number FNS-187494.

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A 模型与超参数

A.1 离散自编码器

我们的离散自编码器基于VQGAN的实现（Esser 等人，2021）。我们去掉了判别器，本质上将VQGAN转变为一个带有额外感知损失的普通VQVAE（Van Den Oord 等人，2017；Johnson 等人，2016；Larsen 等人，2016）。

训练目标如下：

$$\mathcal{L}(E, D, \mathcal{E}) = \|x - D(z)\|_1 + \|\text{sg}(E(x)) - \mathcal{E}(z)\|_2^2 + \|\text{sg}(\mathcal{E}(z)) - E(x)\|_2^2 + \mathcal{L}_{\text{perceptual}}(x, D(z))$$

其中，第一项为重建损失，接下来两项构成承诺损失（其中sg(·)为停止梯度算子），最后一项为感知损失。

表2：编码器/解码器超参数。我们列出了编码器的超参数，解码器采用相同的超参数。

超参数	值
帧尺寸（高，宽）	64×64
层数	4
每层残差块数	2
卷积通道数	64
自注意力层分辨率	8 / 16

表 3：嵌入表超参数。

超参数	取值
词汇表大小 (N)	512
每帧令牌数 (K)	16
令牌嵌入维度 (d)	512

请注意，在真实环境中进行经验收集时，帧仍会通过自编码器，以保持策略的输入分布不变。详见算法1。

A.2 Transformer

我们的自回归Transformer基于minGPT（Karpathy，2020）的实现。它接收一个由 $L(K + 1)$ 个令牌组成的序列，并使用一个针对动作的 $A \times D$ 嵌入表和一个针对帧令牌的 $N \times D$ 嵌入表将其嵌入为 $L(K + 1) \times D$ 维张量。该张量经过 M 个Transformer块处理。我们采用GPT2式块（Radford等，2019），即每个块包含一个带有输入层归一化的自注意力模块，并包裹残差连接，随后是一个带有输入层归一化的逐位置多层感知机，并包裹另一个残差连接。

A MODELS AND HYPERPARAMETERS

A.1 DISCRETE AUTOENCODER

Our discrete autoencoder is based on the implementation of VQGAN (Esser et al., 2021). We removed the discriminator, essentially turning the VQGAN into a vanilla VQVAE (Van Den Oord et al., 2017) with an additional perceptual loss (Johnson et al., 2016; Larsen et al., 2016).

The training objective is the following:

$$\mathcal{L}(E, D, \mathcal{E}) = \|x - D(z)\|_1 + \|\text{sg}(E(x)) - \mathcal{E}(z)\|_2^2 + \|\text{sg}(\mathcal{E}(z)) - E(x)\|_2^2 + \mathcal{L}_{\text{perceptual}}(x, D(z))$$

Here, the first term is the reconstruction loss, the next two terms constitute the commitment loss (where sg(·) is the stop-gradient operator), and the last term is the perceptual loss.

Table 2: Encoder / Decoder hyperparameters. We list the hyperparameters for the encoder, the same ones apply for the decoder.

Hyperparameter	Value
Frame dimensions (h, w)	64×64
Layers	4
Residual blocks per layer	2
Channels in convolutions	64
Self-attention layers at resolution	8 / 16

Table 3: Embedding table hyperparameters.

Hyperparameter	Value
Vocabulary size (N)	512
Tokens per frame (K)	16
Token embedding dimension (d)	512

Note that during experience collection in the real environment, frames still go through the autoencoder to keep the input distribution of the policy unchanged. See Algorithm 1 for details.

A.2 TRANSFORMER

Our autoregressive Transformer is based on the implementation of minGPT (Karpathy, 2020). It takes as input a sequence of $L(K + 1)$ tokens and embeds it into a $L(K + 1) \times D$ tensor using an $A \times D$ embedding table for actions, and a $N \times D$ embedding table for frames tokens. This tensor is forwarded through M Transformer blocks. We use GPT2-like blocks (Radford et al., 2019), i.e. each block consists of a self-attention module with layer normalization of the input, wrapped with a residual connection, followed by a per-position multi-layer perceptron with layer normalization of the input, wrapped with another residual connection.

表4: Transformer超参数

超参数	数值
时间步 (L)	20
嵌入维度 (D)	256
层数 (M)	10
注意力头数	4
权重衰减	0.01
嵌入丢弃率	0.1
注意力丢弃率	0.1
残差丢弃率	0.1

A.3 演员-评论家

演员和评论家网络的权重除了最后一层外均共享。演员-评论家接收一个 $64 \times 64 \times 3$ 帧作为输入，并通过一个卷积块和随后的一个LSTM细胞 (Mnih等人, 2016; Hochreiter和Schmidhuber, 1997; Gers等人, 2000) 进行处理。卷积块由以下相同层重复四次组成: 一个步长为1、填充为1的3x3卷积, 一个ReLU激活函数, 以及一个步长为2的2x2最大池化。LSTM隐藏状态的维度为512。在从给定帧开始想象过程之前, 我们通过20个先前帧进行燃烧期 (Kapturowski等人, 2019) 以初始化隐藏状态。

表 5: 训练循环与共享超参数

ters

超参数	值
轮次	600
# 采集轮次	500
每轮环境步数	200
采集 ϵ -贪婪	0.01
评估采样温度	0.5
自动编码器启动轮次	5
Transformer启动轮次	25
演员-评论家启动轮次	50
自编码器批量大小	256
Transformer 批量大小	64
演员-评论家批量大小	64
每轮训练步数	200
学习率	1e-4
优化器	Adam
Adam β_1	0.9
Adam β_2	0.999
最大梯度范数	10.0

Table 4: Transformer hyperparameters

Hyperparameter	Value
Timesteps (L)	20
Embedding dimension (D)	256
Layers (M)	10
Attention heads	4
Weight decay	0.01
Embedding dropout	0.1
Attention dropout	0.1
Residual dropout	0.1

A.3 ACTOR-CRITIC

The weights of the actor and critic are shared except for the last layer. The actor-critic takes as input a $64 \times 64 \times 3$ frame, and forwards it through a convolutional block followed by an LSTM cell (Mnih et al., 2016; Hochreiter & Schmidhuber, 1997; Gers et al., 2000). The convolutional block consists of the same layer repeated four times: a 3x3 convolution with stride 1 and padding 1, a ReLU activation, and 2x2 max-pooling with stride 2. The dimension of the LSTM hidden state is 512. Before starting the imagination procedure from a given frame, we burn-in (Kapturowski et al., 2019) the 20 previous frames to initialize the hidden state.

Table 5: Training loop & Shared hyperparameters

Hyperparameter	Value
Epochs	600
# Collection epochs	500
Environment steps per epoch	200
Collection epsilon-greedy	0.01
Eval sampling temperature	0.5
Start autoencoder after epochs	5
Start transformer after epochs	25
Start actor-critic after epochs	50
Autoencoder batch size	256
Transformer batch size	64
Actor-critic batch size	64
Training steps per epoch	200
Learning rate	1e-4
Optimizer	Adam
Adam β_1	0.9
Adam β_2	0.999
Max gradient norm	10.0

B 演员-评论家学习目标

我们遵循 Dreamer (Hafner 等人, 2020; 2021) 的做法, 使用通用的 λ -回报 (在偏差和方差之间取得平衡) 作为价值网络的回归目标。给定一条想象轨迹 $\hat{x}_0, \hat{a}_0, \hat{r}_0, \hat{d}_0, \dots, \hat{x}_{H-1}, \hat{a}_{H-1}, \hat{r}_{H-1}, \hat{d}_{H-1}, \hat{x}_H$, λ -回报可按如下方式递归定义:

$$\Lambda_t = \begin{cases} \hat{r}_t + \gamma(1 - \hat{d}_t) \left[(1 - \lambda)V(\hat{x}_{t+1}) + \lambda\Lambda_{t+1} \right] & \text{if } t < H \\ V(\hat{x}_H) & \text{if } t = H \end{cases} \quad (4)$$

价值网络 V 经过训练以最小化 $\mathcal{L}_V$, 即与想象轨迹上 λ -回报的期望平方差。

$$\mathcal{L}_V = \mathbb{E}_\pi \left[\sum_{t=0}^{H-1} (V(\hat{x}_t) - \text{sg}(\Lambda_t))^2 \right] \quad (5)$$

此处, $\text{sg}(\cdot)$ 表示梯度停止操作, 意味着目标是基于梯度优化中的一个常数, 这一做法在文献中已有经典确立 (Mnih 等人, 2015; Hessel 等人, 2018; Hafner 等人, 2020)。

由于在想象MDP中会生成大量轨迹, 我们可以对策略采用直接的强化学习目标, 例如 REINFORCE (Sutton & Barto, 2018)。为了降低REINFORCE梯度的方差, 我们使用价值函数 $V(\hat{x}_t)$ 作为基线 (Sutton & Barto, 2018)。我们还添加了加权熵最大化目标, 以维持足够的探索。演员在想象轨迹上接受训练, 以最小化以下REINFORCE目标:

$$\mathcal{L}_\pi = -\mathbb{E}_\pi \left[\sum_{t=0}^{H-1} \log(\pi(a_t | \hat{x}_{\leq t})) \text{sg}(\Lambda_t - V(\hat{x}_t)) + \eta \mathcal{H}(\pi(a_t | \hat{x}_{\leq t})) \right] \quad (6)$$

表6: 强化学习训练超参数

超参数	数值
想象地平线 (H)	20
γ	0.995
λ	0.95
η	0.001

B ACTOR-CRITIC LEARNING OBJECTIVES

We follow Dreamer (Hafner et al., 2020; 2021) in using the generic λ -return, that balances bias and variance, as the regression target for the value network. Given an imagined trajectory $(\hat{x}_0, a_0, \hat{r}_0, \hat{d}_0, \dots, \hat{x}_{H-1}, a_{H-1}, \hat{r}_{H-1}, \hat{d}_{H-1}, \hat{x}_H)$, the λ -return can be defined recursively as follows:

$$\Lambda_t = \begin{cases} \hat{r}_t + \gamma(1 - \hat{d}_t) \left[(1 - \lambda)V(\hat{x}_{t+1}) + \lambda\Lambda_{t+1} \right] & \text{if } t < H \\ V(\hat{x}_H) & \text{if } t = H \end{cases} \quad (4)$$

The value network V is trained to minimize $\mathcal{L}_V$, the expected squared difference with λ -returns over imagined trajectories.

$$\mathcal{L}_V = \mathbb{E}_\pi \left[\sum_{t=0}^{H-1} (V(\hat{x}_t) - \text{sg}(\Lambda_t))^2 \right] \quad (5)$$

Here, $\text{sg}(\cdot)$ denotes the gradient stopping operation, meaning that the target is a constant in the gradient-based optimization, as classically established in the literature (Mnih et al., 2015; Hessel et al., 2018; Hafner et al., 2020).

As large amounts of trajectories are generated in the imagination MDP, we can use a straightforward reinforcement learning objective for the policy, such as REINFORCE (Sutton & Barto, 2018). To reduce the variance of REINFORCE gradients, we use the value $V(\hat{x}_t)$ as a baseline (Sutton & Barto, 2018). We also add a weighted entropy maximization objective to maintain a sufficient exploration. The actor is trained to minimize the following REINFORCE objective over imagined trajectories:

$$\mathcal{L}_\pi = -\mathbb{E}_\pi \left[\sum_{t=0}^{H-1} \log(\pi(a_t | \hat{x}_{\leq t})) \text{sg}(\Lambda_t - V(\hat{x}_t)) + \eta \mathcal{H}(\pi(a_t | \hat{x}_{\leq t})) \right] \quad (6)$$

Table 6: RL training hyperparameters

Hyperparameter	Value
Imagination horizon (H)	20
γ	0.995
λ	0.95
η	0.001

C 最优性差距

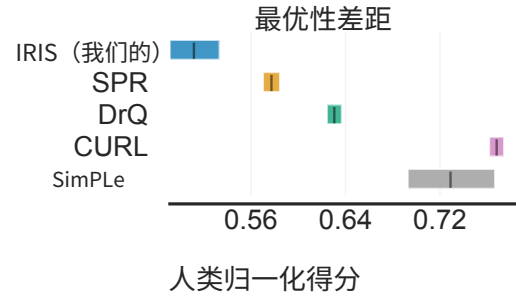


图8：最优性差距，数值越低越好。该指标衡量算法未能达到人类水平分数的程度（Agarwal 等人，2021）。

D IRIS算法

算法1: IRIS**过程** 训练_循环():

对于 轮次 **执行**

 收集_经验(步骤_收集)

对于 步数_世界_模型 **执行**

 更新_世界_模型()

对于 步数_行为 **执行**

 更新_行为()

过程 收集_经验(n):

$x_0 \leftarrow \text{环境重置}()$

for $t = 0$ **到** $n - 1$ **do**

$\hat{x}_t \leftarrow D(E(x_t))$ // 通过离散自编码器前向传输帧

 样本 $a_t \sim \pi(a_t | \hat{x}_t)$

$x_{t+1}, r_t, d_t \leftarrow \text{环境.步进}(a_t)$

if $d_t = 1$ **then**

$x_{t+1} \leftarrow \text{环境重置}()$

$\mathcal{D} \leftarrow \mathcal{D} \cup \{x_t, a_t, r_t, d_t\}_{t=0}^{n-1}$

过程 更新_世界_模型():

 采样 $\{x_t, a_t, r_t, d_t\}_{t=\tau}^{\tau+L-1} \sim \mathcal{D}$

 计算 $z_t := E(x_t)$ 和 $\hat{x}_t := D(z_t)$ 对 $t = \tau, \dots, \tau + L - 1$

 更新 E 和 D

 计算 $p_G(\hat{z}_{t+1}, \hat{r}_t, \hat{d}_t | z_\tau, a_\tau, \dots, z_t, a_t)$ 对 $t = \tau, \dots, \tau + L - 1$

 更新 G

过程 更新_行为():

 采样 $x_0 \sim \mathcal{D}$

$z_0 \leftarrow E(x_0)$

$\hat{x}_0 \leftarrow D(z_0)$

对于 $t = 0$ **到** $H - 1$ **执行**

 样本 $a_t \sim \pi(a_t | \hat{x}_t)$

 样本 $\hat{z}_{t+1}, \hat{r}_t, \hat{d}_t \sim p_G(\hat{z}_{t+1}, \hat{r}_t, \hat{d}_t | z_0, a_0, \dots, \hat{z}_t, a_t)$

$\hat{x}_{t+1} \leftarrow D(\hat{z}_{t+1})$

 计算 $V(\hat{x}_t)$ 针对 $t = 0, \dots, H$

 更新 π 和 V

C OPTIMALITY GAP

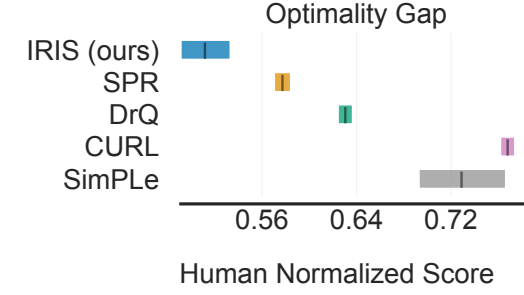


Figure 8: Optimality gap, lower is better. The amount by which the algorithm fails to reach a human-level score (Agarwal et al., 2021).

D IRIS ALGORITHM

Algorithm 1: IRIS**Procedure** training_loop():

for epochs **do**

 collect_experience(steps_collect)

for steps_world_model **do**

 update_world_model()

for steps_behavior **do**

 update_behavior()

Procedure collect_experience(n):

$x_0 \leftarrow \text{env.reset}()$

for $t = 0$ **to** $n - 1$ **do**

$\hat{x}_t \leftarrow D(E(x_t))$ // forward frame through discrete autoencoder

 Sample $a_t \sim \pi(a_t | \hat{x}_t)$

$x_{t+1}, r_t, d_t \leftarrow \text{env.step}(a_t)$

if $d_t = 1$ **then**

$x_{t+1} \leftarrow \text{env.reset}()$

$\mathcal{D} \leftarrow \mathcal{D} \cup \{x_t, a_t, r_t, d_t\}_{t=0}^{n-1}$

Procedure update_world_model():

 Sample $\{x_t, a_t, r_t, d_t\}_{t=\tau}^{\tau+L-1} \sim \mathcal{D}$

 Compute $z_t := E(x_t)$ and $\hat{x}_t := D(z_t)$ for $t = \tau, \dots, \tau + L - 1$

 Update E and D

 Compute $p_G(\hat{z}_{t+1}, \hat{r}_t, \hat{d}_t | z_\tau, a_\tau, \dots, z_t, a_t)$ for $t = \tau, \dots, \tau + L - 1$

 Update G

Procedure update_behavior():

 Sample $x_0 \sim \mathcal{D}$

$z_0 \leftarrow E(x_0)$

$\hat{x}_0 \leftarrow D(z_0)$

for $t = 0$ **to** $H - 1$ **do**

 Sample $a_t \sim \pi(a_t | \hat{x}_t)$

 Sample $\hat{z}_{t+1}, \hat{r}_t, \hat{d}_t \sim p_G(\hat{z}_{t+1}, \hat{r}_t, \hat{d}_t | z_0, a_0, \dots, \hat{z}_t, a_t)$

$\hat{x}_{t+1} \leftarrow D(\hat{z}_{t+1})$

 Compute $V(\hat{x}_t)$ for $t = 0, \dots, H$

 Update π and V

编码器：使用可变令牌数量对帧进行自编码

Transformer 的序列长度由用于编码单个帧的令牌数量以及记忆中的时间步数量决定。增加每帧令牌数可以提高重建质量， 尽管这需要更多的计算和内存资源。

这种权衡在视觉复杂且配置可能性较多的游戏中尤为重要， 因为离散自编码器仅使用 16 个令牌时难以正确编码帧。例如， 图 9 显示， 在*Alien*中将每帧令牌数增加到 64 时， 离散自编码器能够正确重建玩家、敌人和奖励。

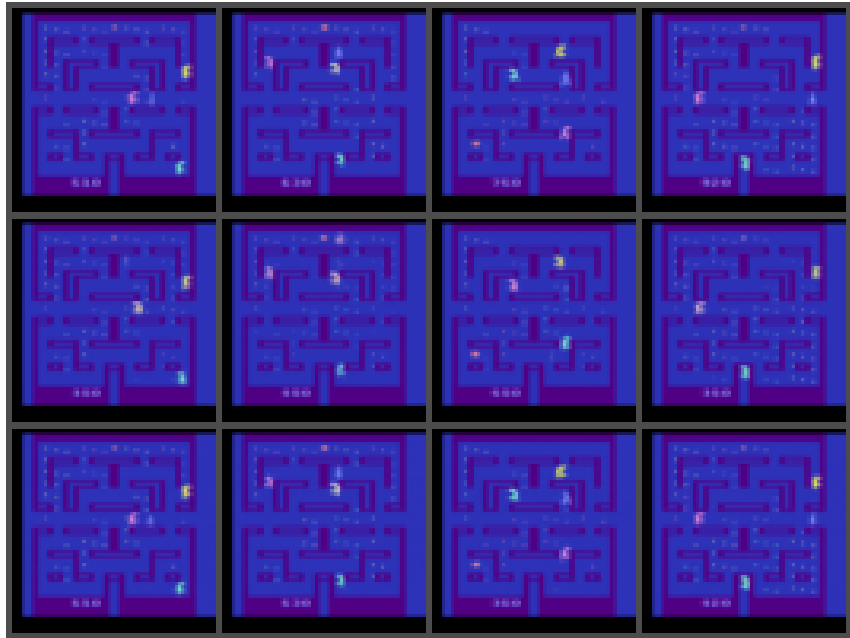


图 9: *Alien*中每帧令牌数与重建质量之间的权衡。每列显示来自真实环境的一帧 64×64 图像（顶部）、使用 16 个令牌的离散编码重建结果（中间）以及使用 64 个令牌的离散编码重建结果（底部）。在*Alien*中， 玩家是深蓝色角色， 敌人是大型彩色精灵。当每帧使用 16 个令牌时， 自编码器经常丢失玩家、 切换颜色并错误放置奖励。而增加令牌数量后， 它能够正确重建帧。

表7展示了IRIS在三种游戏中使用每帧64个令牌训练时的最终性能。有趣的是， 尽管世界模型更加准确， 但在*Alien*中的性能仅略有提升（+36%）。这一观察表明， *Alien*提出了一个困难的强化学习问题， 其他基线在该游戏中的低性能也证明了这一点。相反， 对于Asterix（+121%）和BankHeist（+432%）， IRIS从每帧更多令牌中获益显著。

表7：使用每帧64个令牌而非16个令牌时， 在*Alien*、Asterix和BankHeist上的回报表现。

Game	随机	人类	SimPLe	CURL	DrQ	SPR	IRIS (16个令牌)	IRIS (64个令牌)
Alien	227.8	7127.7	616.9	711.0	865.2	841.9	420.0	570.0
Asterix	210.0	8503.3	1128.3	567.2	763.6	962.5	853.6	1890.4
BankHeist	14.2	753.1	34.2	65.3	232.9	345.4	53.1	282.5

E AUTOENCODING FRAMES WITH VARYING AMOUNTS OF TOKENS

The sequence length of the Transformer is determined by the number of tokens used to encode a single frame and the number of timesteps in memory. Increasing the number of tokens per frame results in better reconstructions, although it requires more compute and memory.

This tradeoff is particularly important in visually challenging games with a high number of possible configurations, where the discrete autoencoder struggles to properly encode frames with only 16 tokens. For instance, Figure 9 shows that, when increasing the number of tokens per frame to 64 in *Alien*, the discrete autoencoder correctly reconstructs the player, its enemies, and rewards.

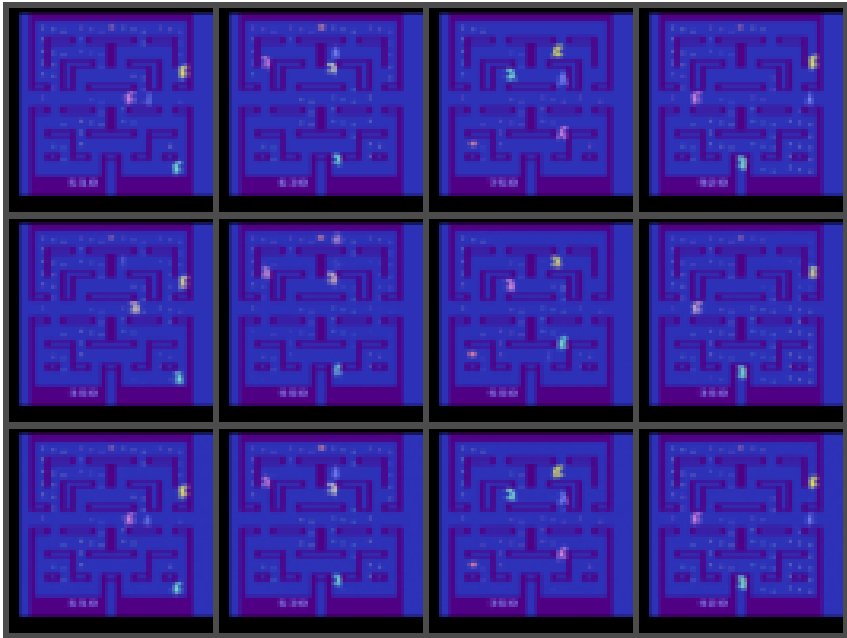


Figure 9: Tradeoff between the number of tokens per frame and reconstructions quality in *Alien*. Each column displays a 64×64 frame from the real environment (top), its reconstruction with a discrete encoding of 16 tokens (center), and its reconstruction with a discrete encoding of 64 tokens (bottom). In *Alien*, the player is the dark blue character, and the enemies are the large colored sprites. With 16 tokens per frame, the autoencoder often erases the player, switches colors, and misplaces rewards. When increasing the amount of tokens, it properly reconstructs the frame.

Table 7 displays the final performance of IRIS trained with 64 tokens per frame in three games. Interestingly, even though the world model is more accurate, the performance in *Alien* only increases marginally (+36%). This observation suggests that *Alien* poses a hard reinforcement learning problem, as evidenced by the low performance of other baselines in that game. On the contrary, IRIS greatly benefits from having more tokens per frame for Asterix (+121%) and BankHeist (+432%).

Table 7: Returns on *Alien*, Asterix, and BankHeist with 64 tokens per frame instead of 16.

Game	Random	Human	SimPLe	CURL	DrQ	SPR	IRIS (16 tokens)	IRIS (64 tokens)
Alien	227.8	7127.7	616.9	711.0	865.2	841.9	420.0	570.0
Asterix	210.0	8503.3	1128.3	567.2	763.6	962.5	853.6	1890.4
BankHeist	14.2	753.1	34.2	65.3	232.9	345.4	53.1	282.5

F 超越样本效率设定

IRIS可以通过增加用于编码帧的令牌数量、增加模型容量、在每个环境步骤中执行更多优化步骤或使用更多数据进行扩展。在本实验中，我们通过将环境步骤从10万增加到1000万来研究数据缩放特性。然而，为了将训练时间控制在我们的计算资源范围内，我们将每个环境步骤的优化步骤比例从1:1降低到1:50。因此，本实验在10万帧时的结果会比论文中报告的结果更差。

表8：将环境步骤从10万步增加到1000万步。

Game	随机	人类	IRIS (10万步)	IRIS (10M)
Alien	227.8	7127.7	420.0	1003.1
Amidar	5.8	1719.5	143.0	213.4
Assault	222.4	742.0	1524.4	9355.6
Asterix	210.0	8503.3	853.6	6861.0
BankHeist	14.2	753.1	53.1	921.6
BattleZone	2360.0	37187.5	13074.0	34562.5
Boxing	0.1	12.1	70.1	98.0
Breakout	1.7	30.5	83.7	493.9
ChopperCommand	811.0	7387.8	1565.0	9814.0
CrazyClimber	10780.5	35829.4	59324.2	111068.8
DemonAttack	152.1	1971.0	2034.4	96218.6
Freeway	0.0	29.6	31.1	34.0
Frostbite	65.2	4334.7	259.1	290.3
Gopher	257.6	2412.5	2236.1	97370.6
Hero	1027.0	30826.4	7037.4	19212.0
Jamesbond	29.0	302.8	462.7	5534.4
Kangaroo	52.0	3035.0	838.2	1793.8
Krull	1598.0	2665.5	6616.4	7344.0
KungFuMaster	258.5	22736.3	21759.8	39643.8
MsPacman	307.3	6951.6	999.1	1233.0
Pong	-20.7	14.6	14.6	21.0
PrivateEye	24.9	69571.3	100.0	100.0
Qbert	163.9	13455.0	745.7	4012.1
RoadRunner	11.5	7845.0	9614.6	30609.4
Seaquest	68.4	42054.7	661.3	1815.0
UpNDown	533.4	11693.2	3546.2	114690.1
#超人类 (↑)	0	N/A	10	15
平均值 (↑)	0.000	1.000	1.046	7.488
中位数 (↑)	0.000	1.000	0.289	1.207
IQM (↑)	0.000	1.000	0.501	2.239
最优性差距 (↓)	1.000	0.000	0.512	0.282

表8表明，将环境步骤从10万增加到1000万，能显著提升大多数游戏的性能，这证明了IRIS可以扩展到样本高效区域之外。在某些游戏中，更多数据仅带来边际改善，这很可能是因为困难探索问题或视觉挑战领域的存在，而这些情况可以通过增加编码帧的令牌数量（附录E）来改善。

F BEYOND THE SAMPLE-EFFICIENT SETTING

IRIS can be scaled up by increasing the number of tokens used to encode frames, adding capacity to the model, taking more optimization steps per environment steps, or using more data. In this experiment, we investigate data scaling properties by increasing the number of environment steps from 100k to 10M. However, to maintain a training time within our computational resources, we lower the ratio of optimization steps per environment steps from 1:1 to 1:50. As a consequence, the results of this experiment at 100k frames would be worse than those reported in the paper.

Table 8: Increasing the number of environment steps from 100k to 10M.

Game	Random	Human	IRIS (100k)	IRIS (10M)
Alien	227.8	7127.7	420.0	1003.1
Amidar	5.8	1719.5	143.0	213.4
Assault	222.4	742.0	1524.4	9355.6
Asterix	210.0	8503.3	853.6	6861.0
BankHeist	14.2	753.1	53.1	921.6
BattleZone	2360.0	37187.5	13074.0	34562.5
Boxing	0.1	12.1	70.1	98.0
Breakout	1.7	30.5	83.7	493.9
ChopperCommand	811.0	7387.8	1565.0	9814.0
CrazyClimber	10780.5	35829.4	59324.2	111068.8
DemonAttack	152.1	1971.0	2034.4	96218.6
Freeway	0.0	29.6	31.1	34.0
Frostbite	65.2	4334.7	259.1	290.3
Gopher	257.6	2412.5	2236.1	97370.6
Hero	1027.0	30826.4	7037.4	19212.0
Jamesbond	29.0	302.8	462.7	5534.4
Kangaroo	52.0	3035.0	838.2	1793.8
Krull	1598.0	2665.5	6616.4	7344.0
KungFuMaster	258.5	22736.3	21759.8	39643.8
MsPacman	307.3	6951.6	999.1	1233.0
Pong	-20.7	14.6	14.6	21.0
PrivateEye	24.9	69571.3	100.0	100.0
Qbert	163.9	13455.0	745.7	4012.1
RoadRunner	11.5	7845.0	9614.6	30609.4
Seaquest	68.4	42054.7	661.3	1815.0
UpNDown	533.4	11693.2	3546.2	114690.1
#Superhuman (↑)	0	N/A	10	15
Mean (↑)	0.000	1.000	1.046	7.488
Median (↑)	0.000	1.000	0.289	1.207
IQM (↑)	0.000	1.000	0.501	2.239
Optimality Gap (↓)	1.000	0.000	0.512	0.282

Table 8 illustrates that increasing the number of environment steps from 100k to 10M drastically improves performance for most games, providing evidence that IRIS could be scaled up beyond the sample-efficient regime. On some games, more data only yields marginal improvements, most likely due to hard exploration problems or visually challenging domains that would benefit from a higher number of tokens to encode frames (Appendix E).

G 计算资源

对于每个 Atari 环境，我们使用 5 种不同的随机种子反复训练 IRIS。我们的实验使用了 8 块英伟达 A100 40GB GPU。当两个 Atari 环境在同一块 GPU 上运行时，训练大约需要 7 天，平均每个环境耗时 3.5 天。

SimPLe (Kaiser 等人, 2020) 是唯一涉及想象中学习的基线，它在单个环境中使用 P100 GPU 训练了 3 周。至于 SPR (Schwarzer 等人, 2021)，作为没有前瞻搜索的最强基线，它在使用 P100 GPU 的情况下仅需 4.6 小时即可完成训练，速度显著较快。

关于使用前瞻搜索的基线，MuZero (Schrittwieser 等人, 2020) 最初使用 40 个 TPU 训练 12 小时才能在单个 Atari 环境中进行训练。Ye 等人 (2021) 使用 4 块 RTX 3090 GPU 在 7 小时内训练了 EfficientZero 及其对 MuZero 的重新实现。EfficientZero 的实现依赖于一个分布式基础设施，其中 CPU 和 GPU 线程并行运行，以及一个 C++/Cython 实现的蒙特卡洛树搜索。相比之下，IRIS 和不使用前瞻搜索的基线则依赖于简单的单 GPU / 单 CPU 实现。

H 在 Freeway 中的探索

Freeway 的奖励函数是稀疏的，因为智能体只有在完全穿过马路时才会获得奖励。此外，与汽车碰撞会将其推下，阻止其顺利向上攀爬高速公路。这对新初始化的智能体构成了探索问题，因为随机策略在 10 万帧预算内几乎肯定永远无法获得非零奖励。



图10：一局 *Freeway* 游戏。汽车会将玩家撞下，使得随机策略极难穿过马路并获得奖励。

解决这个问题的方法其实很简单，只需要在部分时间段内对向上动作进行过采样。大多数 Atari 10 万基准通过 ϵ -贪心调度和 argmax 动作选择来修复该问题，此时网络配置会在某个时间点使向上动作被高度偏好。在这项工作中，我们选择了更简单的策略：使用固定 ϵ -贪心参数并从策略中进行采样。但是，我们将 Freeway 的采样温度从 1 降低到了 0.01，以避免在训练初期出现不利于学习的随机游走。因此，一旦通过探索获得了最初的几次奖励，IRIS 就能够将稀疏奖励函数内化到其世界模型中。

G COMPUTATIONAL RESOURCES

For each Atari environment, we repeatedly trained IRIS with 5 different random seeds. We ran our experiments with 8 Nvidia A100 40GB GPUs. With two Atari environments running on the same GPU, training takes around 7 days, resulting in an average of 3.5 days per environment.

SimPLe (Kaiser et al., 2020), the only baseline that involves learning in imagination, trains for 3 weeks with a P100 GPU on a single environment. As for SPR (Schwarzer et al., 2021), the strongest baseline without lookahead search, it trains notably fast in 4.6 hours with a P100 GPU.

Regarding baselines with lookahead search, MuZero (Schrittwieser et al., 2020) originally used 40 TPUs for 12 hours to train in a single Atari environment. Ye et al. (2021) train both EfficientZero and their reimplementation of MuZero in 7 hours with 4 RTX 3090 GPUs. EfficientZero’s implementation relies on a distributed infrastructure with CPU and GPU threads running in parallel, and a C++/Cython implementation of MCTS. By contrast, IRIS and the baselines without lookahead search rely on straightforward single GPU / single CPU implementations.

H EXPLORATION IN FREEWAY

The reward function in Freeway is sparse since the agent is only rewarded when it completely crosses the road. In addition, bumping into cars will drag it down, preventing it from smoothly ascending the highway. This poses an exploration problem for newly initialized agents because a random policy will almost surely never obtain a non-zero reward with a 100k frames budget.



Figure 10: A game of *Freeway*. Cars will bump the player down, making it very unlikely to cross the road and be rewarded for random policies.

The solution to this problem is actually straightforward and simply requires stretches of time when the UP action is oversampled. Most Atari 100k baselines fix the issue with epsilon-greedy schedules and argmax action selection, where at some point the network configuration will be such that the UP action is heavily favored. In this work, we opted for the simpler strategy of having a fixed epsilon-greedy parameter and sampling from the policy. However, we lowered the sampling temperature from 1 to 0.01 for Freeway, in order to avoid random walks that would not be conducive to learning in the early stages of training. As a consequence, once it received its first few rewards through exploration, IRIS was able to internalize the sparse reward function in its world model.